THE MIKROTIK PLAYBOOK

From First Packet 2 Production

VERSION 1.0

A Practical Guide to Real ISP Networking

MD. HASAN

PREFACE

I have organized this book based on MikroTik in a simple and practical way. The main focus is not just on basic topics, but on real configurations that are used in daily work, especially in ISP environments.

Everything is explained in an easy way so that you can understand quickly and apply it without confusion. Step-by-step examples are included using both Winbox and CLI, so you can follow along and practice on your own.

This book is arranged from basic to advanced topics, so even if you are a beginner, you can start from the beginning and gradually build your skills.

My goal with this book is to make MikroTik learning easier and more practical. After going through this book, you should be able to configure devices and handle real network problems with confidence.

এই বইটি আমি MikroTik-এর উপর সহজ এবং বাস্তবমুখীভাবে সাজিয়েছি। এখানে শুধু বেসিক বিষয় নয়, বরং দৈনন্দিন কাজের বাস্তব কনফিগারেশন—বিশেষ করে ISP পরিবেশে ব্যবহৃত বিষয়গুলো—আলোচনা করা হয়েছে।

সবকিছু সহজভাবে বোঝানো হয়েছে, যাতে আপনি দ্রুত বুঝতে পারেন এবং কোনো বিভ্রান্তি ছাড়াই বাস্তবে প্রয়োগ করতে পারেন। Winbox এবং CLI—দুটোরই step-by-step উদাহরণ দেওয়া আছে, যাতে আপনি নিজে নিজে প্র্যাকটিস করতে পারেন।

বইটি বেসিক থেকে অ্যাডভান্স পর্যন্ত ধারাবাহিকভাবে সাজানো হয়েছে। তাই আপনি যদি একজন বিগিনার হন, তাহলে শুরু থেকে ধাপে ধাপে শিখে নিজের দক্ষতা বাড়াতে পারবেন।

এই বইটির মূল উদ্দেশ্য হলো MikroTik শেখাকে সহজ এবং কার্যকর করা। বইটি শেষ করার পর আপনি আত্মবিশ্বাসের সাথে ডিভাইস কনফিগার করতে এবং বাস্তব নেটওয়ার্ক সমস্যাগুলো সমাধান করতে পারবেন।

MD. HASAN

Heading 1: Basic MikroTik Configuration

Topic 1.1 – MikroTik Configuration using Dynamic IP (with Gateway, DNS & NAT)

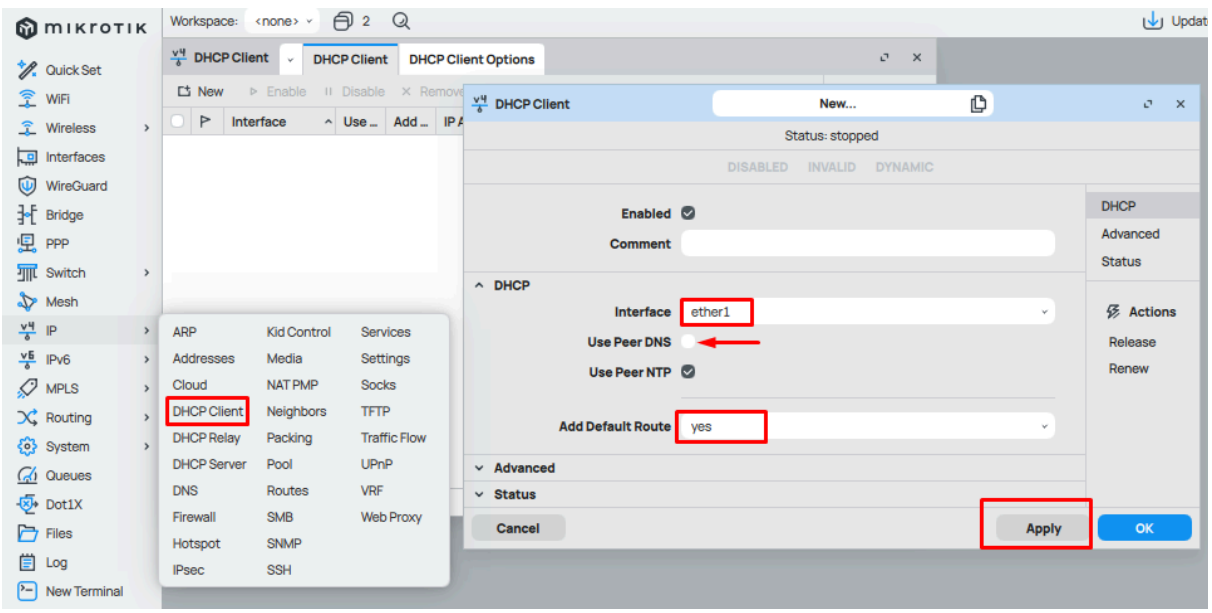
Step 1: Configure WAN Interface (Dynamic IP)

CLI:

```
/ip dhcp-client add interface=ether1 use-peer-dns=no use-peer-ntp=yes add-default-route=yes
```

Winbox:

- **Go to IP → DHCP Client → Add (+)**
- **Select Interface: ether1**
- **Check Add Default Route**
- **Uncheck Use Peer DNS if you want custom DNS**



Note that: When using Dynamic IP, the gateway comes automatically from the ISP via DHCP. You don't need to configure it manually—just check /ip route print to see it.

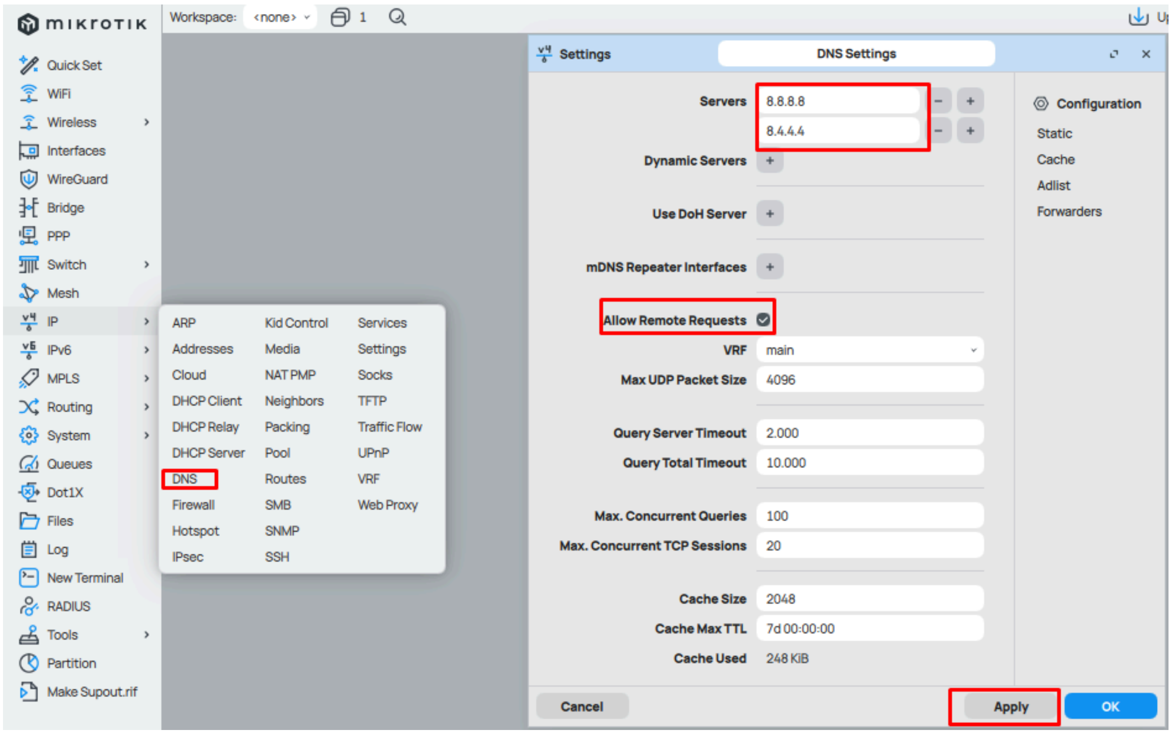
Step 2: Configure DNS

CLI:

```
/ip dns set servers=8.8.8.8,8.8.4.4 allow-remote-requests=yes
```

Winbox:

Go to IP → DNS
Enter 8.8.8.8, 8.8.4.4
Enable Allow Remote Requests



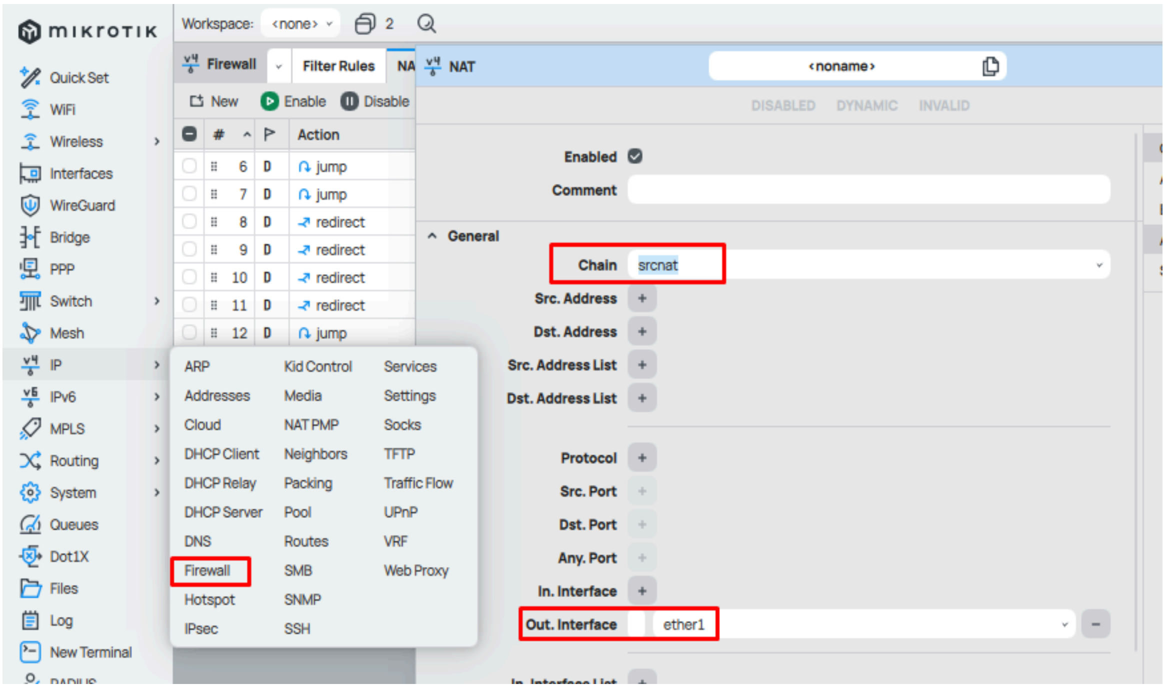
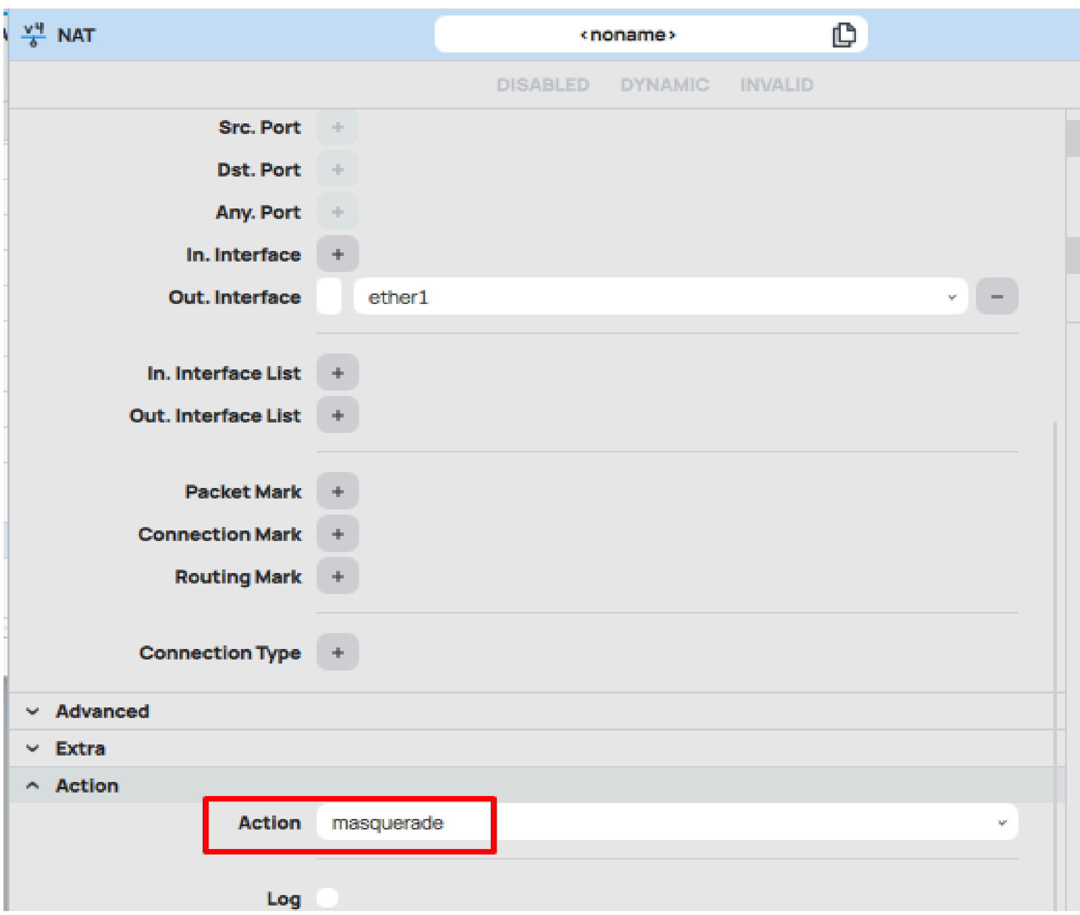
Step 3: Set NAT (Masquerade)

CLI:

```
/ip firewall nat add chain=srcnat action=masquerade out-interface=ether1
```

Winbox:

- **Go to IP → Firewall → NAT → Add (+)**
- **Chain: srcnat**
- **Out Interface: ether1**
- **Action: masquerade**


MikroTik
Workspace: <none> 2
Quick Set
WiFi
Wireless
Interfaces
WireGuard
Bridge
PPP
Switch
Mesh
IP
IPv6
MPLS
Routing
System
Queues
Dot1X
Files
Log
New Terminal
Firewall
Filter Rules
New
Enable
Disable
Action
6 D jump
7 D jump
8 D redirect
9 D redirect
10 D redirect
11 D redirect
12 D jump
ARP
Addresses
Cloud
DHCP Client
DHCP Relay
DHCP Server
DNS
Firewall
Hotspot
IPsec
Kid Control
Media
NAT PMP
Neighbors
Packing
Pool
Routes
SMB
SNMP
SSH
Services
Settings
Socks
TFTP
Traffic Flow
UPnP
VRF
Web Proxy
NAT
‹noname›
DISABLED DYNAMIC INVALID
Enabled
Comment
General
Chain srcnat
Src. Address
Dst. Address
Src. Address List
Dst. Address List
Protocol
Src. Port
Dst. Port
Any. Port
In. Interface
Out. Interface ether1

NAT
‹noname›
DISABLED DYNAMIC INVALID
Src. Port
Dst. Port
Any. Port
In. Interface
Out. Interface ether1
In. Interface List
Out. Interface List
Packet Mark
Connection Mark
Routing Mark
Connection Type
Advanced
Extra
Action
Action masquerade
Log

Topic 1.2 – MikroTik Configuration using Static IP (with Gateway, DNS & NAT, Without Bridging)

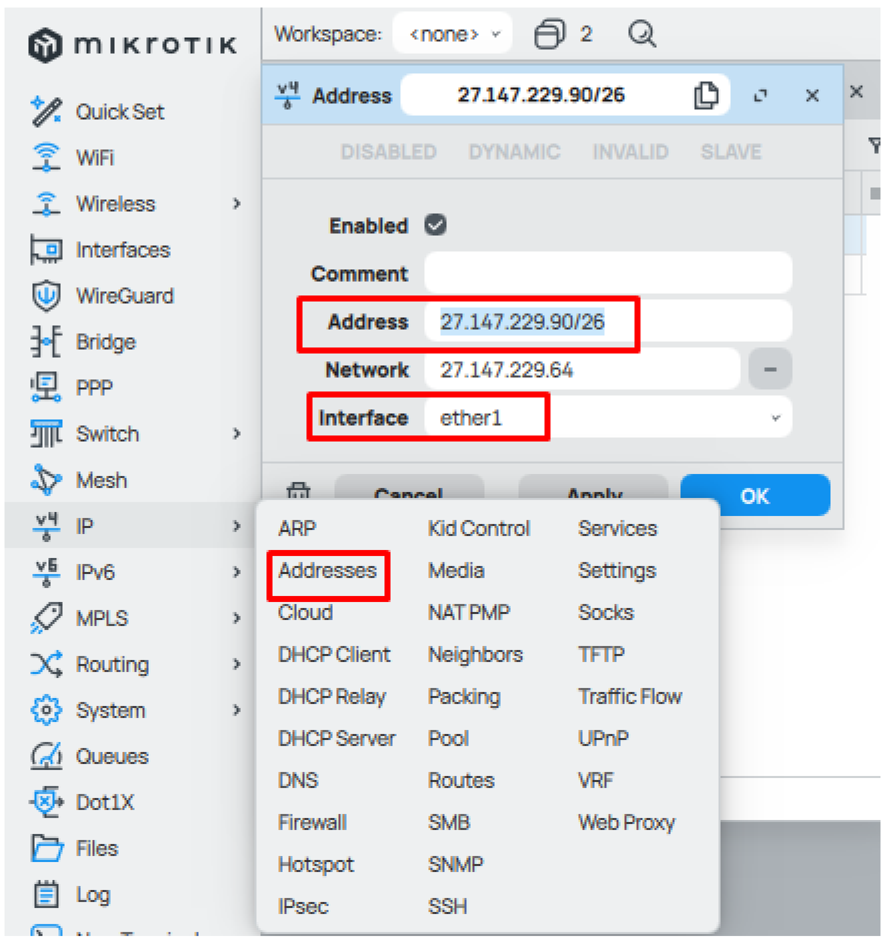
Step 1: Configure WAN Interface (Static IP)

CLI:

```
/ip address add address=27.147.229.90/26 interface=ether1
```

Winbox:

- Go to IP → Addresses → Add (+)
- Address: 27.147.229.90/26
- Interface: ether1



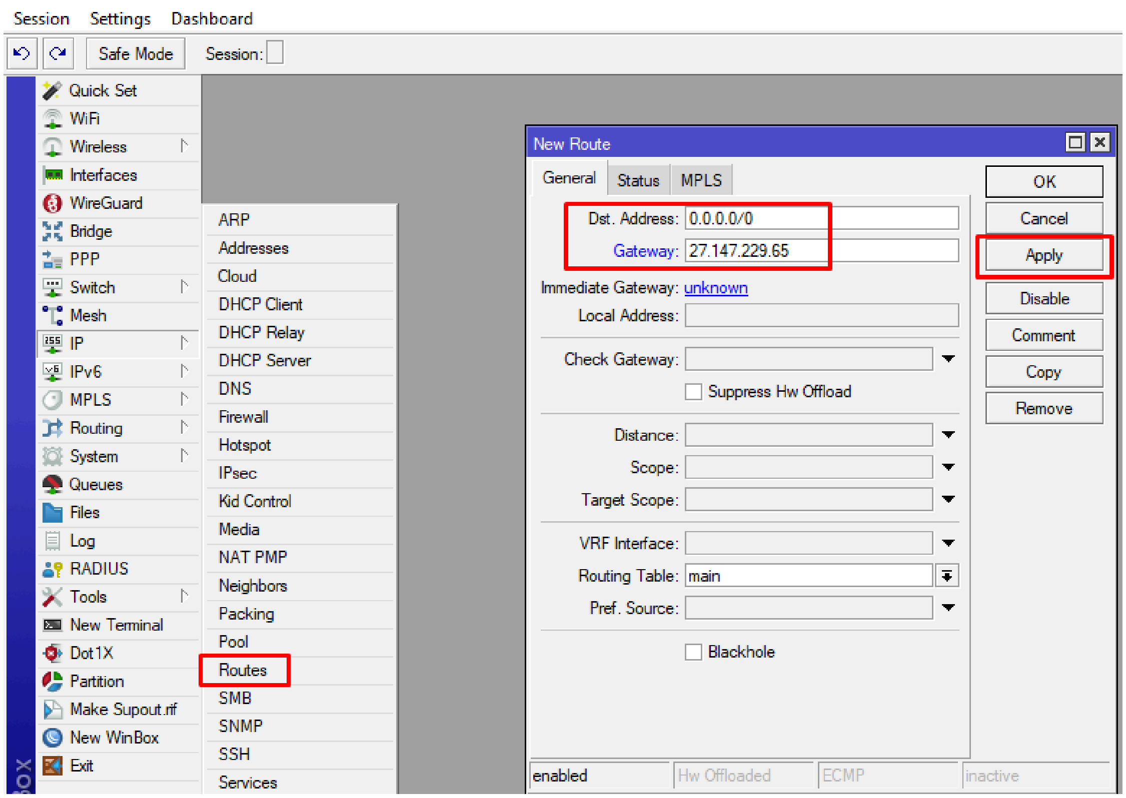
Step 2: Configure Gateway

CLI:

```
/ip route add gateway=27.147.229.65
```

Winbox:

- **Go to IP → Routes → Add (+)**
- **Gateway:** 27.147.229.65



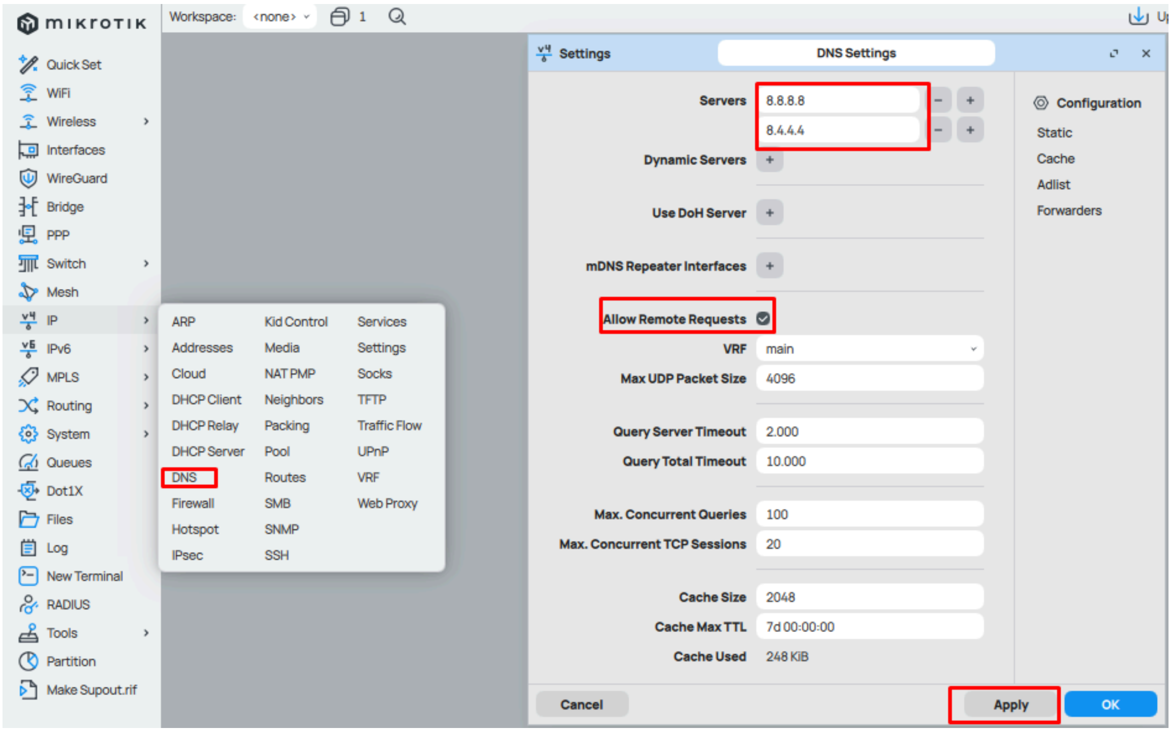
Step 3: Configure DNS

CLI:

```
/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes
```

Winbox:

- **Go to IP → DNS**
- **Enter** 8.8.8.8, 1.1.1.1
- **Enable Allow Remote Requests**



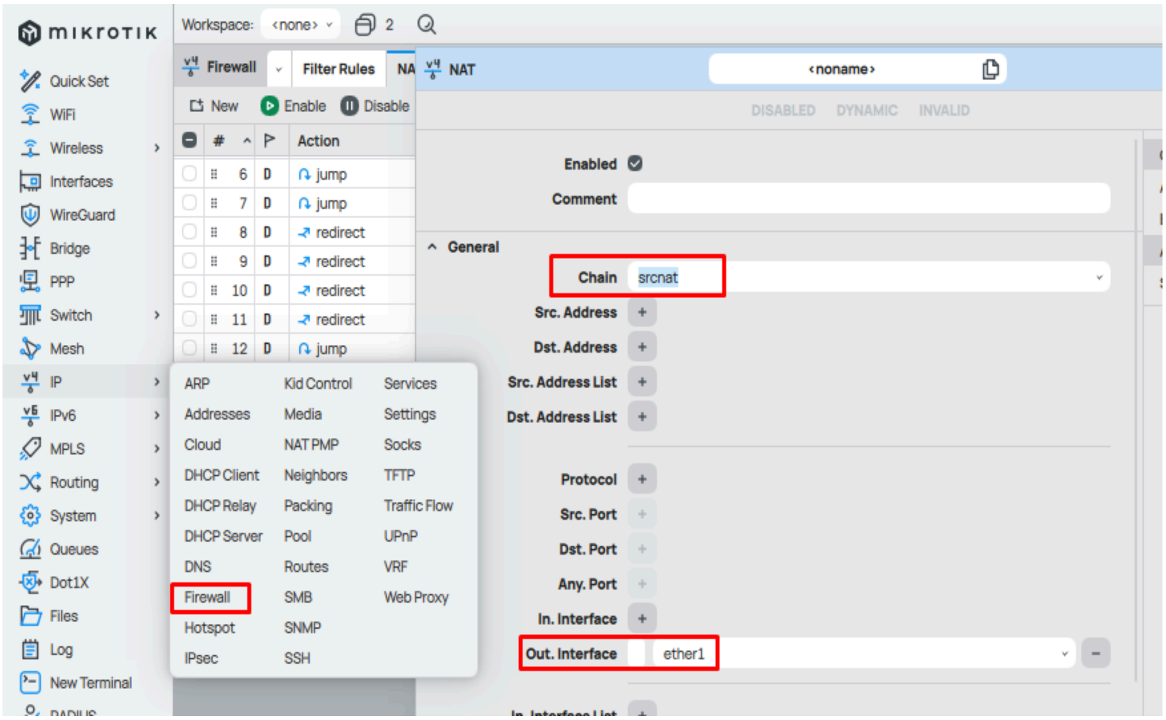
Step 4: Configure NAT (Masquerade)

CLI:

```
/ip firewall nat add chain=srcnat action=masquerade out-interface=ether1
```

Winbox:

- Go to IP → Firewall → NAT → Add (+)
- Chain: srcnat
- Out Interface: ether1
- Action: masquerade



Topic 1.3 – Bridging Two Ports and Configuring with Dynamic & Static IP (with Gateway, DNS & NAT)

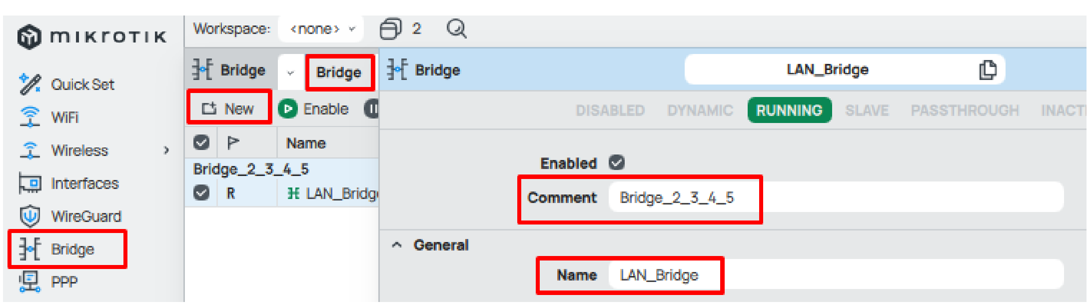
Step 1: Create Bridge and Add LAN Ports

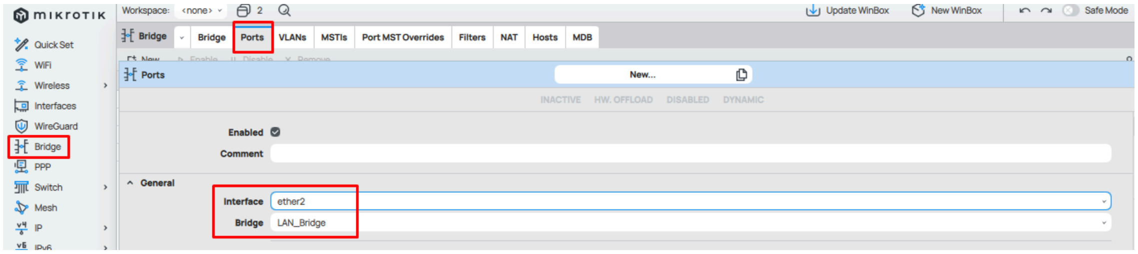
CLI:

```
/interface bridge add name=bridge1
/interface bridge port add bridge=bridge1 interface=ether2
/interface bridge port add bridge=bridge1 interface=ether3
```

Winbox:

- Go to Bridge → Bridge → Add (+) → Name bridge1
- Go to Bridge → Ports → Add (+)
- Add ether2, ether3 to bridge1

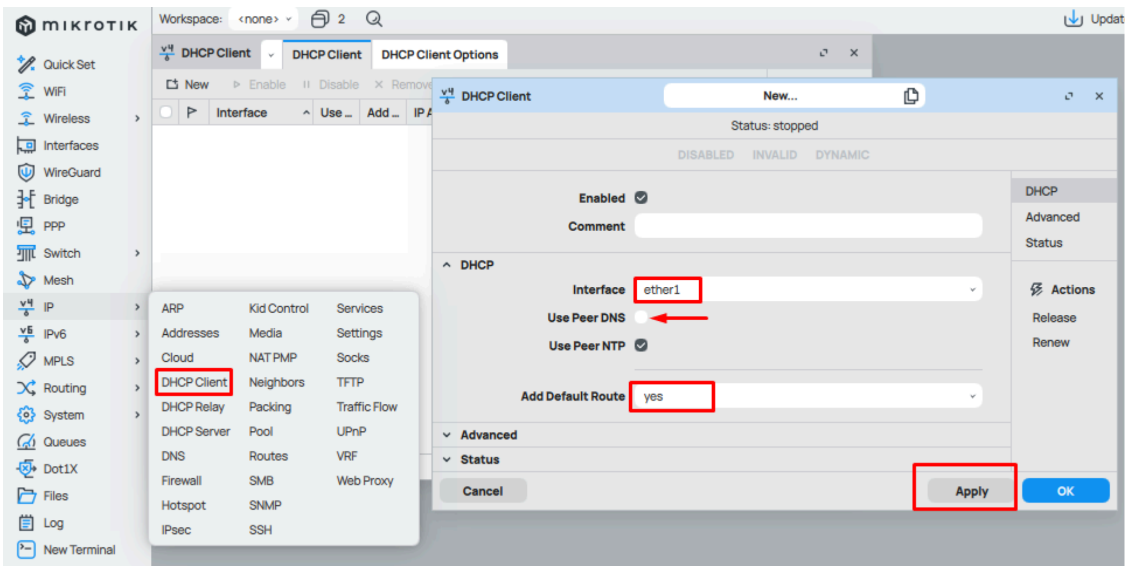




Step 2: Configure WAN (Dynamic IP or Static IP)

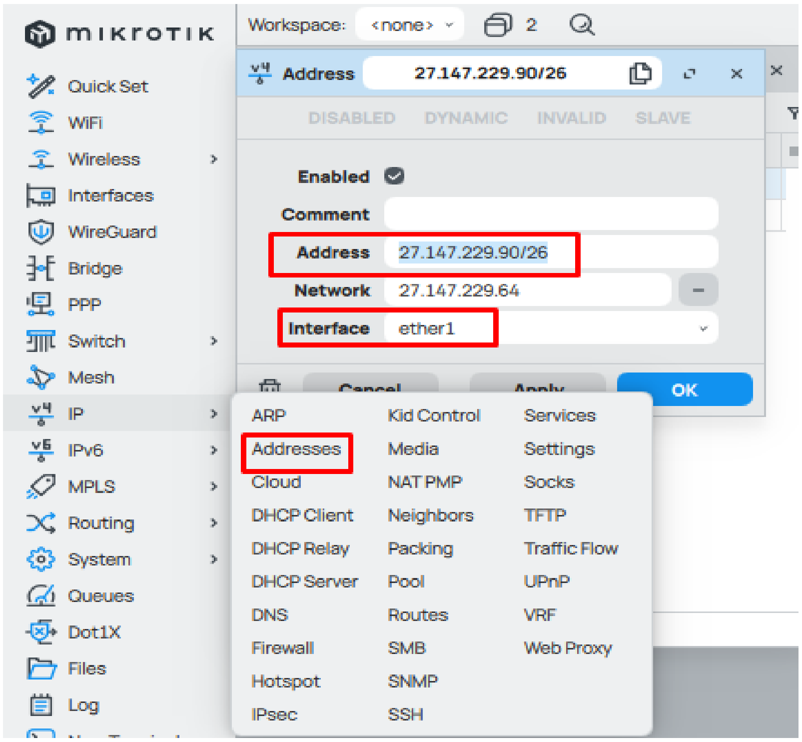
Dynamic:

```
/ip dhcp-client add interface=ether1 use-peer-dns=no add-default-route=yes
```

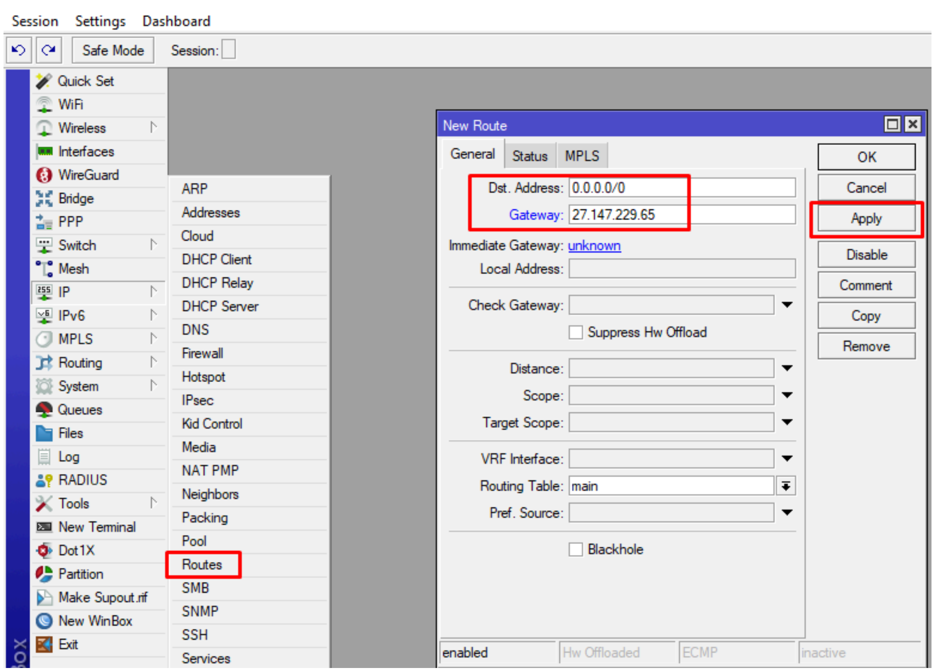


Static:

```
/ip address add address=27.147.229.90/26 interface=ether1
```



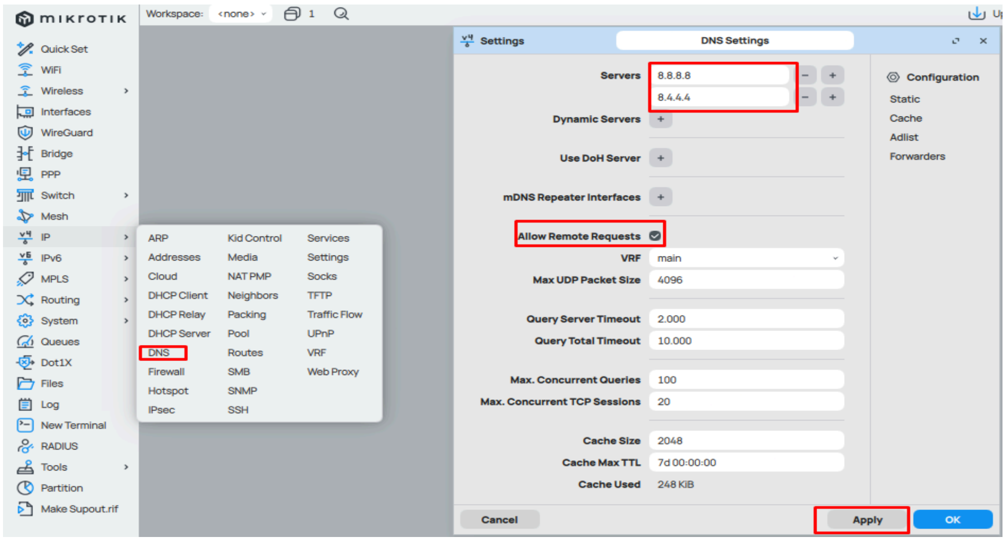
/ip route add gateway=27.147.229.65



Step 3: Configure DNS

CLI:

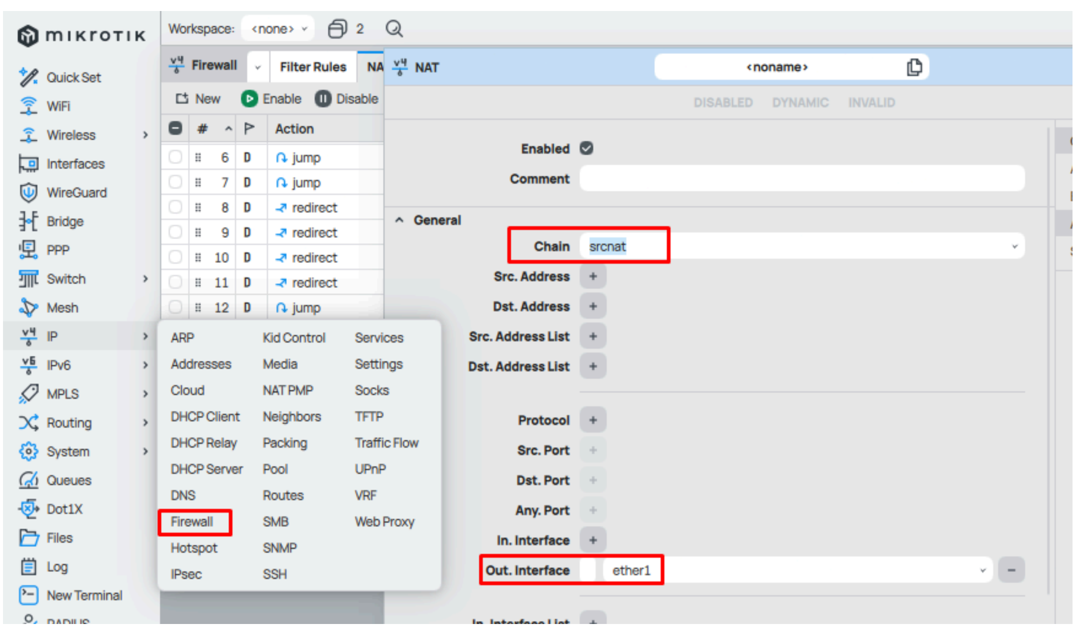
/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes

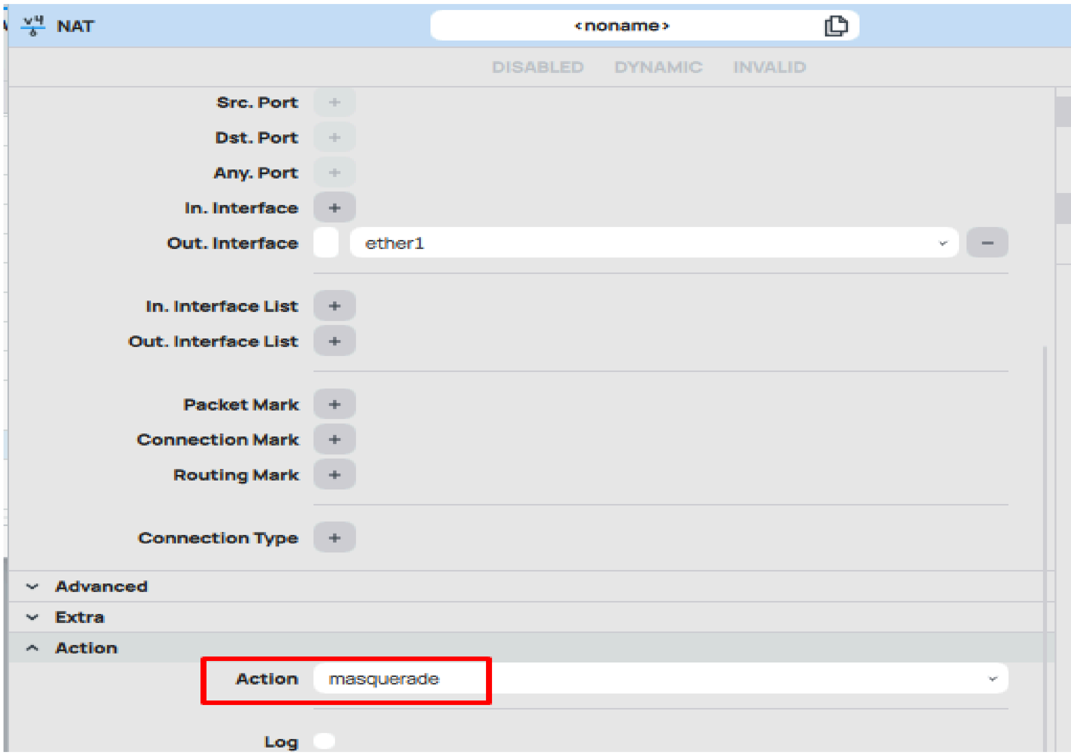


Step 4: Configure NAT (Masquerade)

CLI:

/ip firewall nat add chain=srcnat action=masquerade out-interface=ether1




NAT
<noname>
DISABLED
DYNAMIC
INVALID
Src. Port
Dst. Port
Any. Port
In. Interface
Out. Interface
ether1
In. Interface List
Out. Interface List
Packet Mark
Connection Mark
Routing Mark
Connection Type
Advanced
Extra
Action
Action
masquerade
Log

Heading 2: LAN & DHCP Setup

Topic 2.1 – Assigning LAN IP to Bridge/LAN Interface

Step 1: Assign IP to LAN (Bridge or Single Interface)

CLI:

```
/ip address add address=192.168.10.1/24 interface=bridge1
```

(If not using a bridge, assign to ether2 directly)

```
/ip address add address=192.168.10.1/24 interface=ether2
```

Winbox:

- **Go to IP → Addresses → Add (+)**
- **Address: 192.168.10.1/24**
- **Interface: LAN_Bridge (or ether2)**

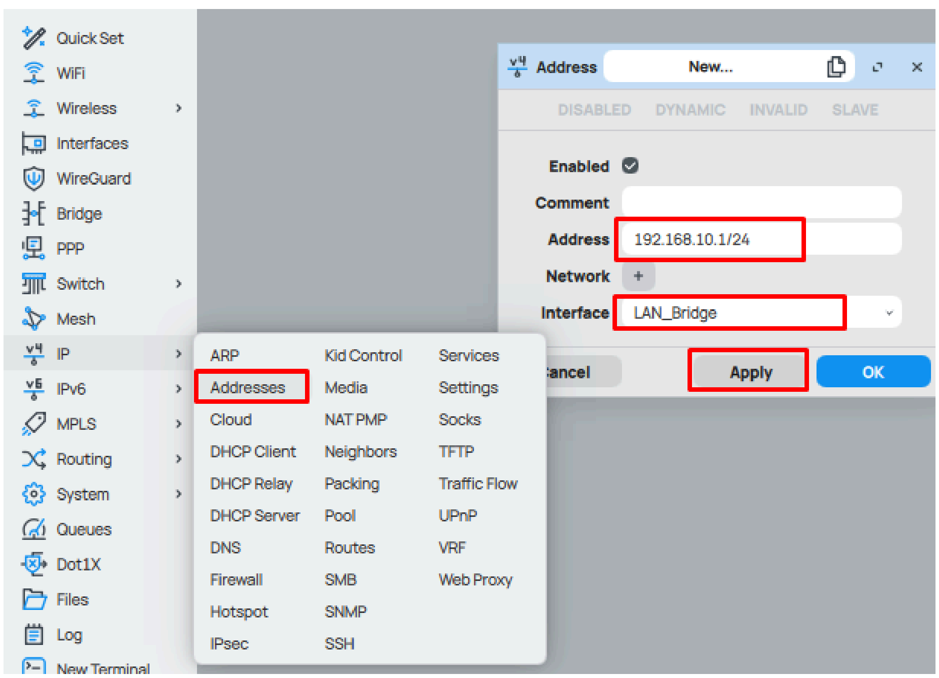


Image - Assign IP to LAN (Bridge or Single Interface)

Topic 2.2 – Configure DHCP Server for LAN

Step 1: Create IP Pool

CLI:

/ip pool add name=LAN-Pool ranges=192.168.10.10-192.168.10.254

Winbox:

- Go to IP → Pool → Add (+)
- Name: LAN-Pool
- Range: 192.168.10.10-192.168.10.254

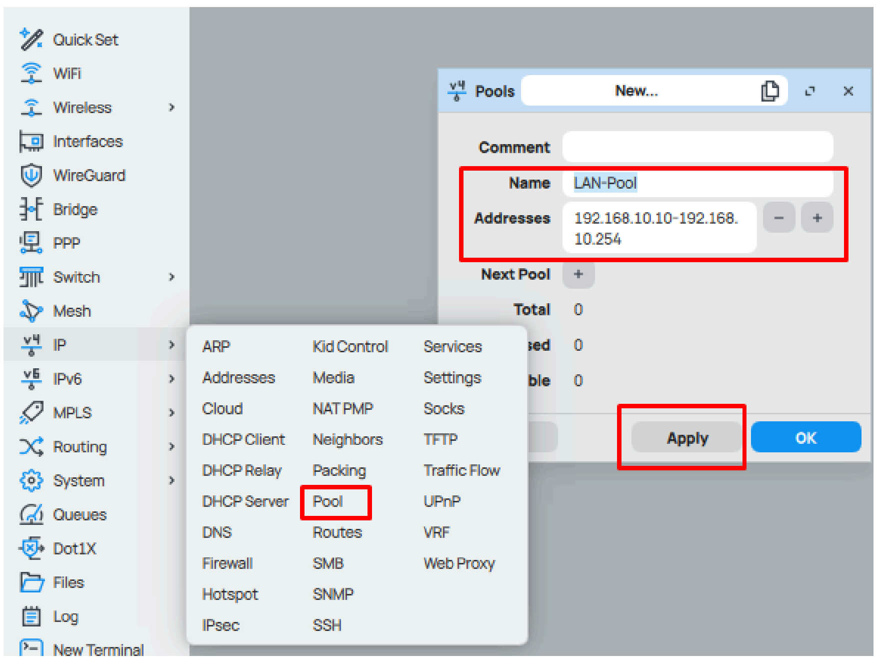


Image - Create IP Pool

Step 2: Configure DHCP Network

CLI:

/ip dhcp-server network add address=192.168.10.0/24 gateway=192.168.10.1 dns-server=8.8.8.8,1.1.1.1

Winbox:

- Go to IP → DHCP Server → Networks → Add (+)
- Address: 192.168.10.0/24
- Gateway: 192.168.10.1
- DNS Servers: 8.8.8.8,1.1.1.1

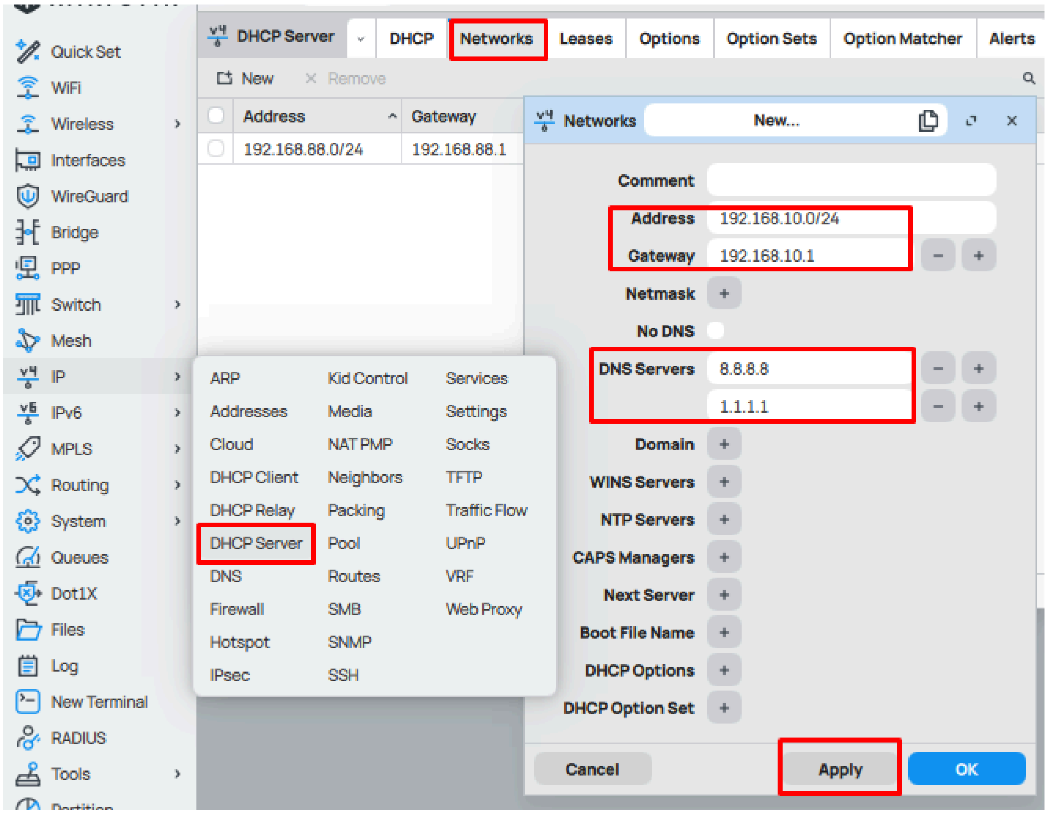


Image - Configure DHCP Network

Step 3: Add and Enable DHCP Server

CLI:

```
/ip dhcp-server add name=LAN-DHCP interface=bridge1 address-pool=LAN-Pool disabled=no
```

(If not using bridge, use interface=ether2)

Winbox:

- Go to IP → DHCP Server → DHCP Setup
- Select Interface: bridge1 (or ether2)
- Set Pool, Gateway, and DNS as prompted

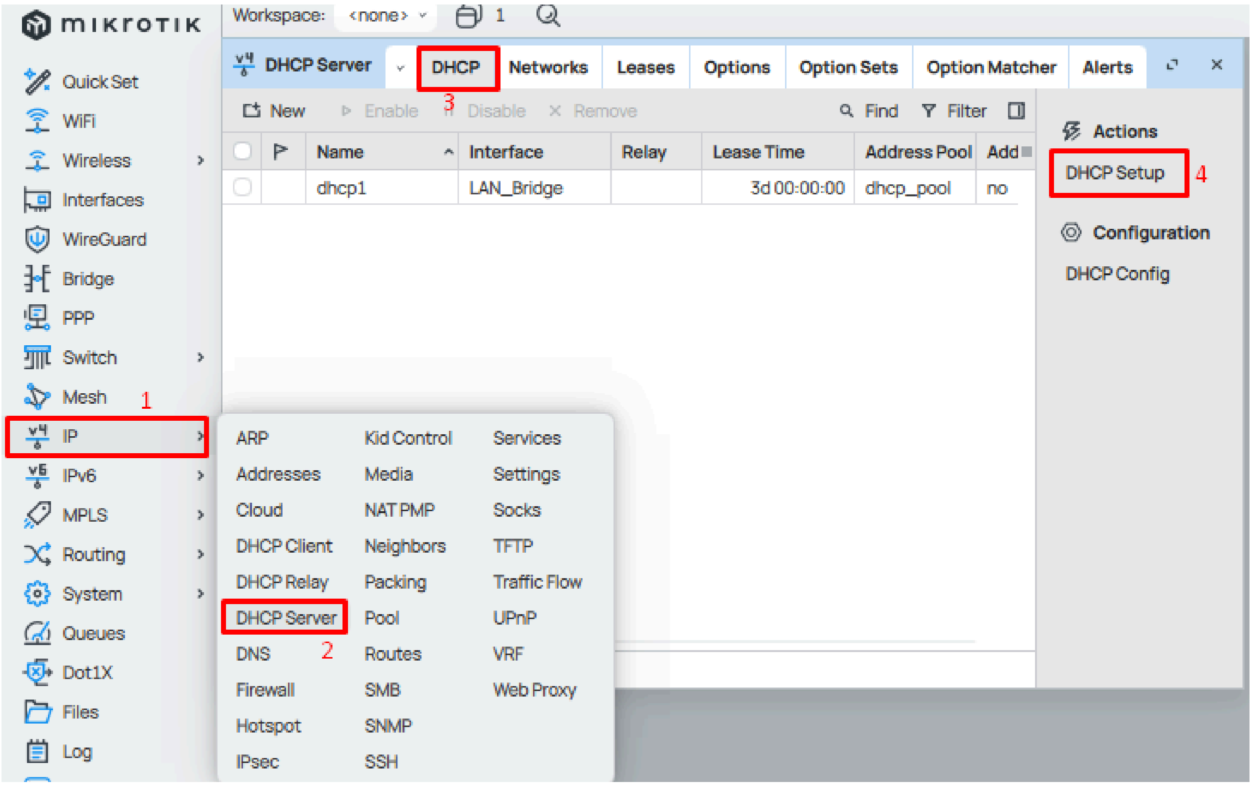


Image: Add DHCP Server

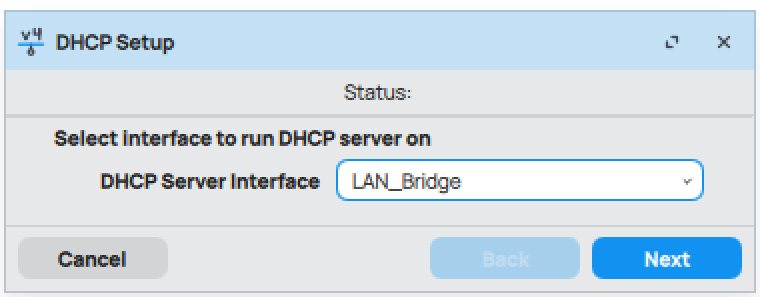


Image: DHCP Setup 1

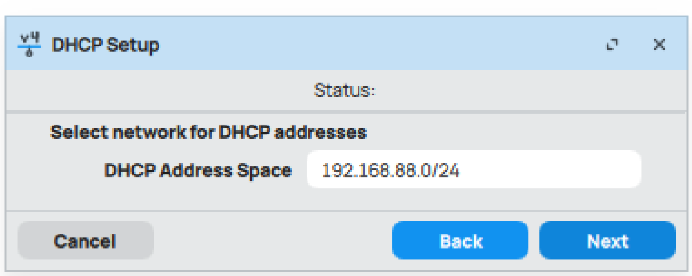


Image: DHCP Setup 2

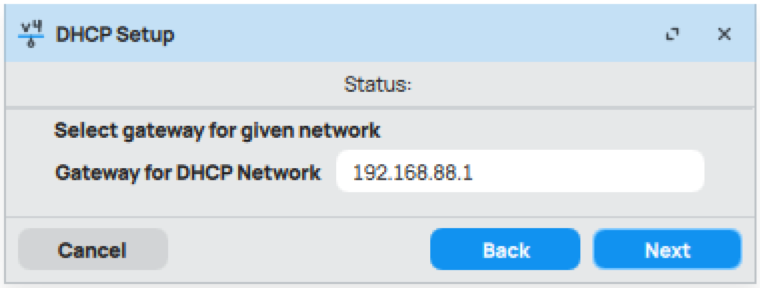


Image: DHCP Setup 3

Topic 2.3 – Test LAN Connectivity

CLI:

/ping 8.8.8.8

Winbox:

- Go to Tools → Ping
- Enter 8.8.8.8

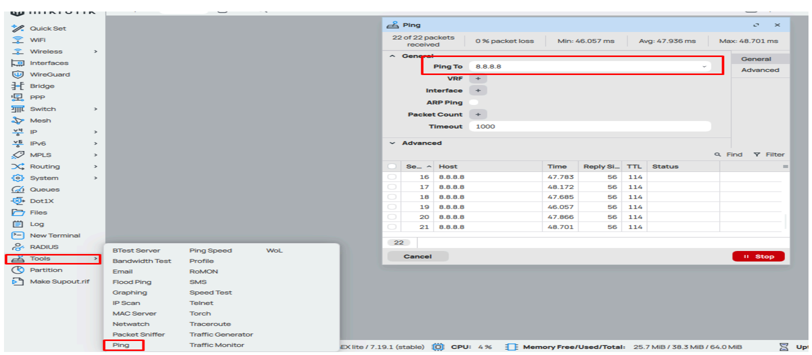


Image: Test Connectivity

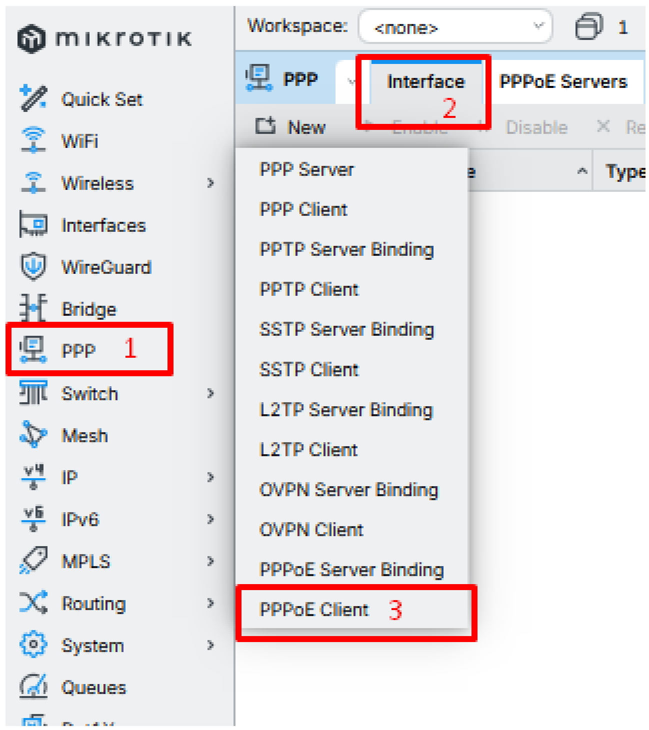
If successful, LAN clients should have automatic IP via DHCP and Internet access via NAT.

Heading 3: PPPoE Configuration by DHCP and Static IP

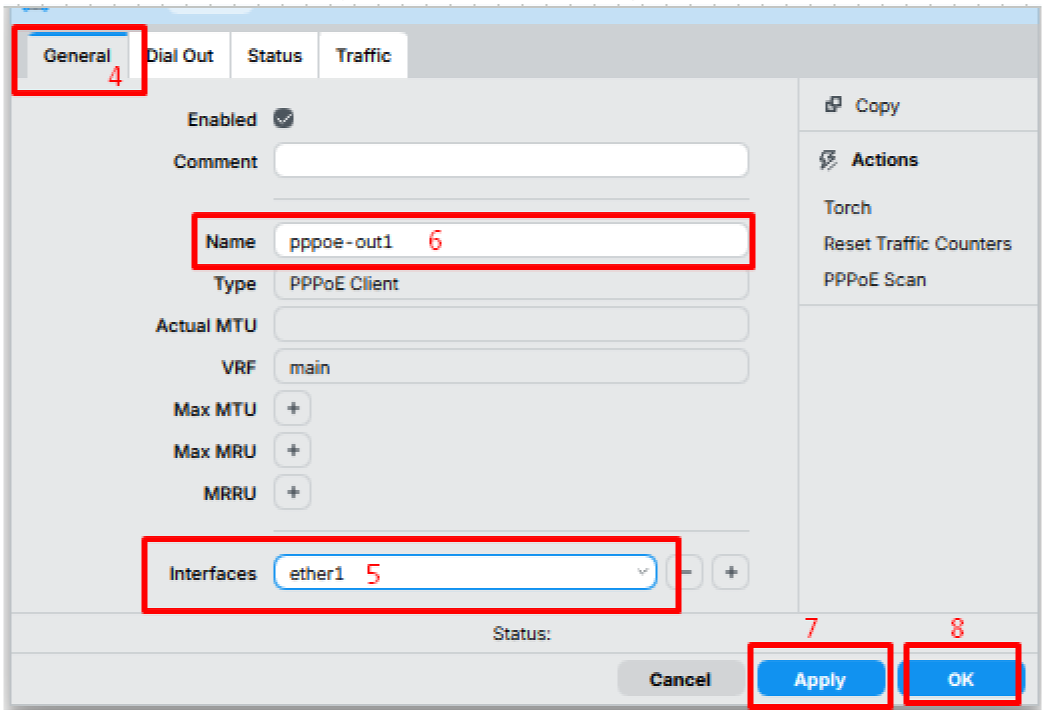
Topic 3.1 – PPPoE with DHCP (Dynamic IP)

Winbox:

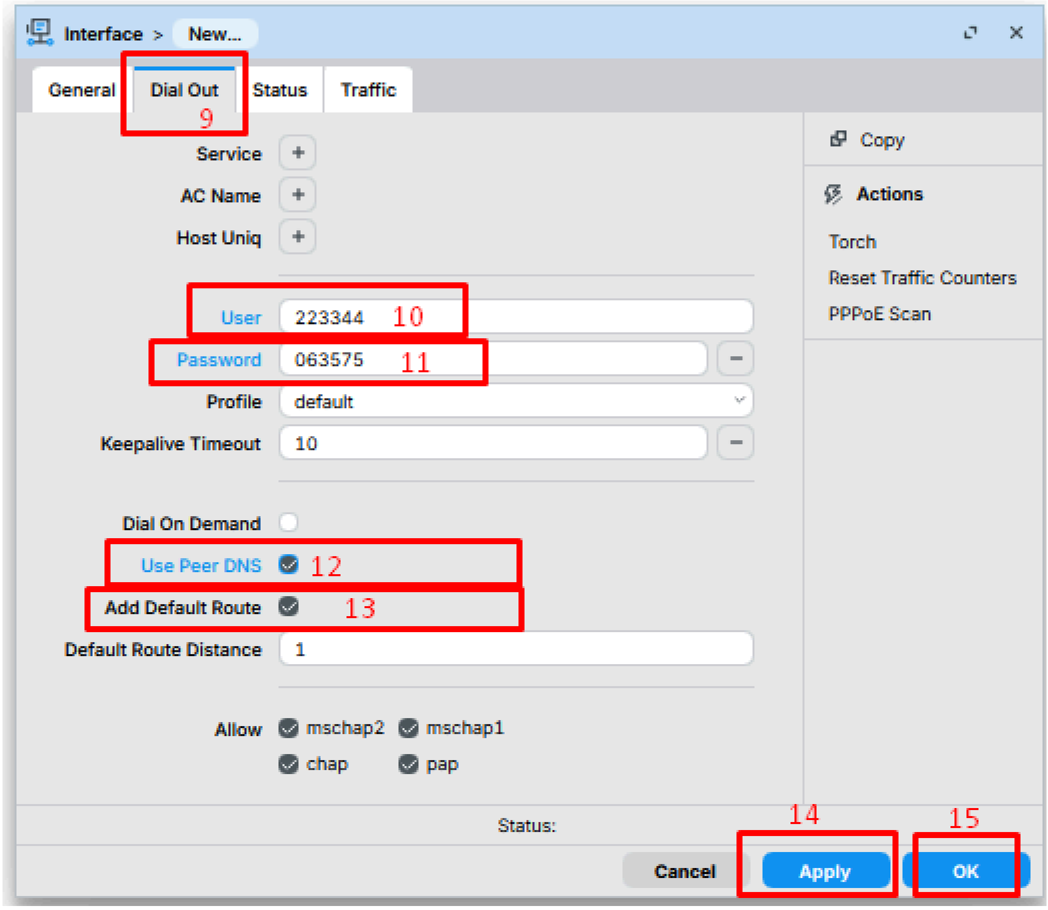
1. **Open Winbox → Go to Interfaces → PPP.**
2. **Click "+" → Select PPPoE Client.**



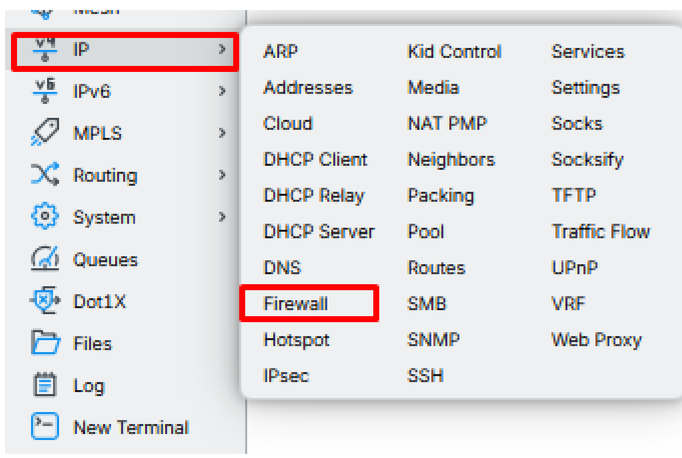
3. In the General tab, choose your WAN interface (e.g., ether1).



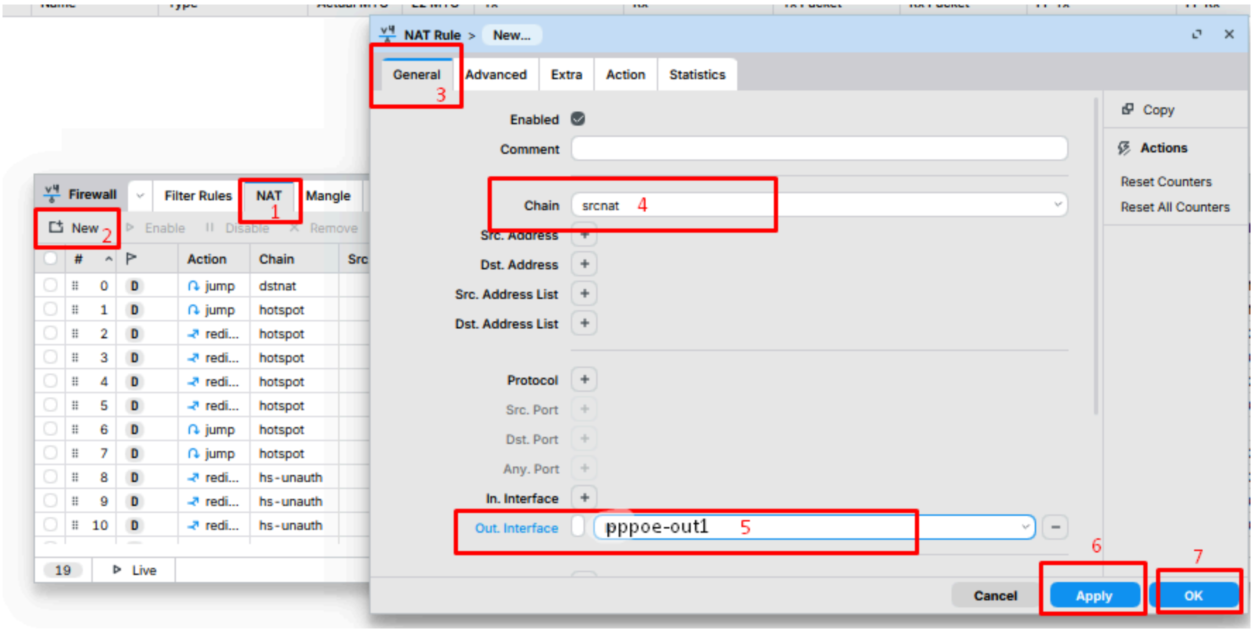
4. In Dial Out tab, enter:
   - Username = 223344
   - Password = 063575
   - Check Use Peer DNS
   - Check Add Default Route
5. Click Apply → OK.



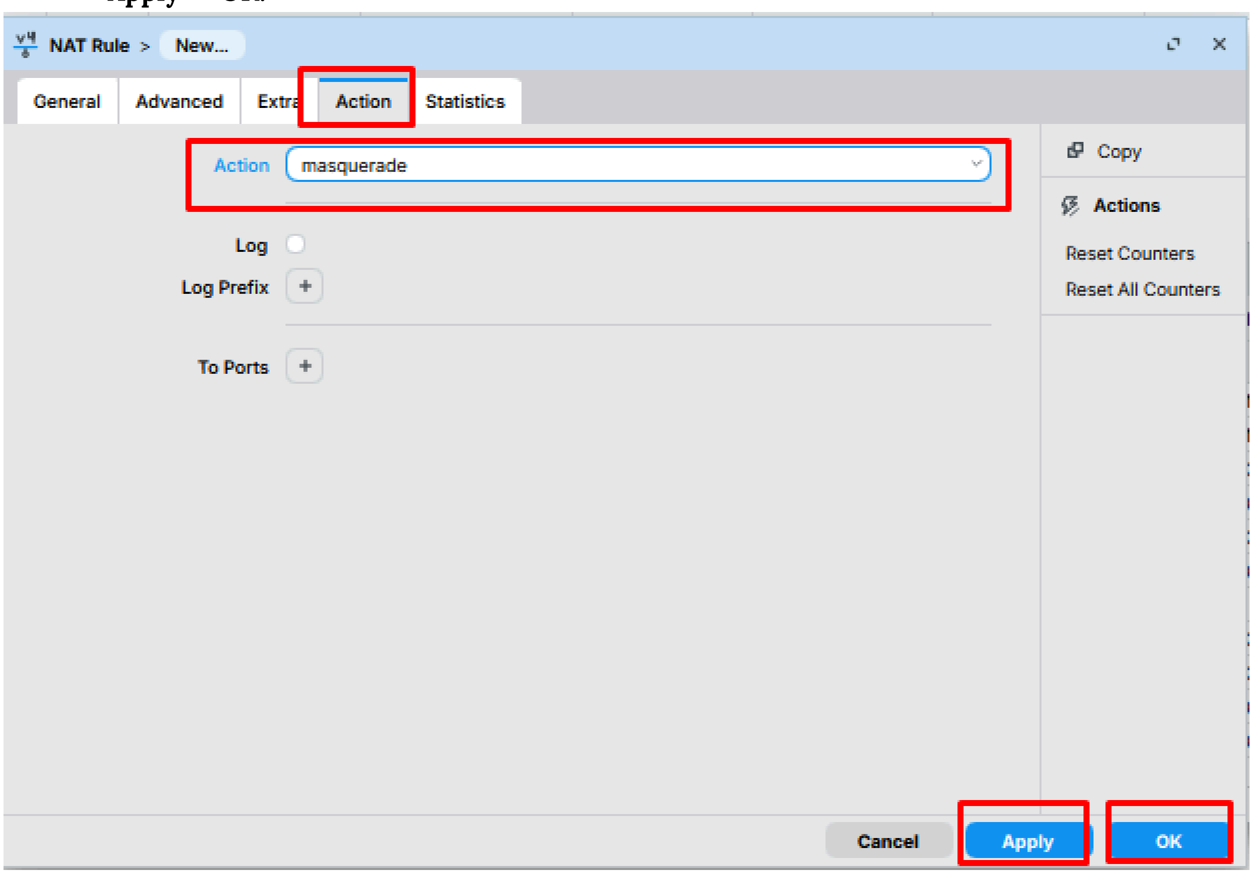
6. Status should show "Connected" and IP will be assigned automatically.
7. Go to IP → Firewall → NAT → +.



- Chain = srcnat
- Out. Interface = pppoe-out1



- Action = masquerade
- Apply → OK.



8. Gateway will be assigned automatically by ISP (check in IP → Routes).

CLI:

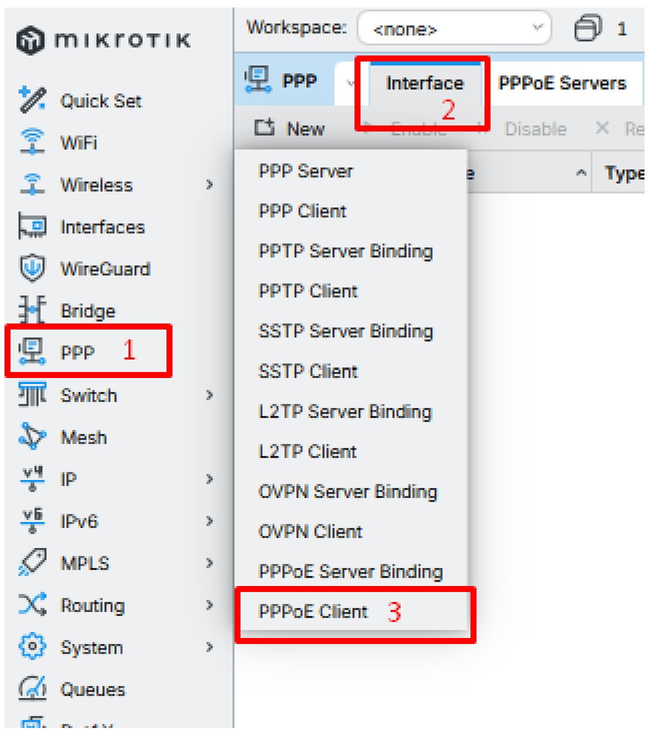
```
/interface pppoe-client add name=pppoe-out1 interface=ether1 user=236403
password=063575 use-peer-dns=yes add-default-route=yes disabled=no

/ip firewall nat add chain=srcnat out-interface=pppoe-out1 action=masquerade

/ip route print   # shows dynamic gateway from ISP
```

Topic 3.2 – PPPoE with Static IP

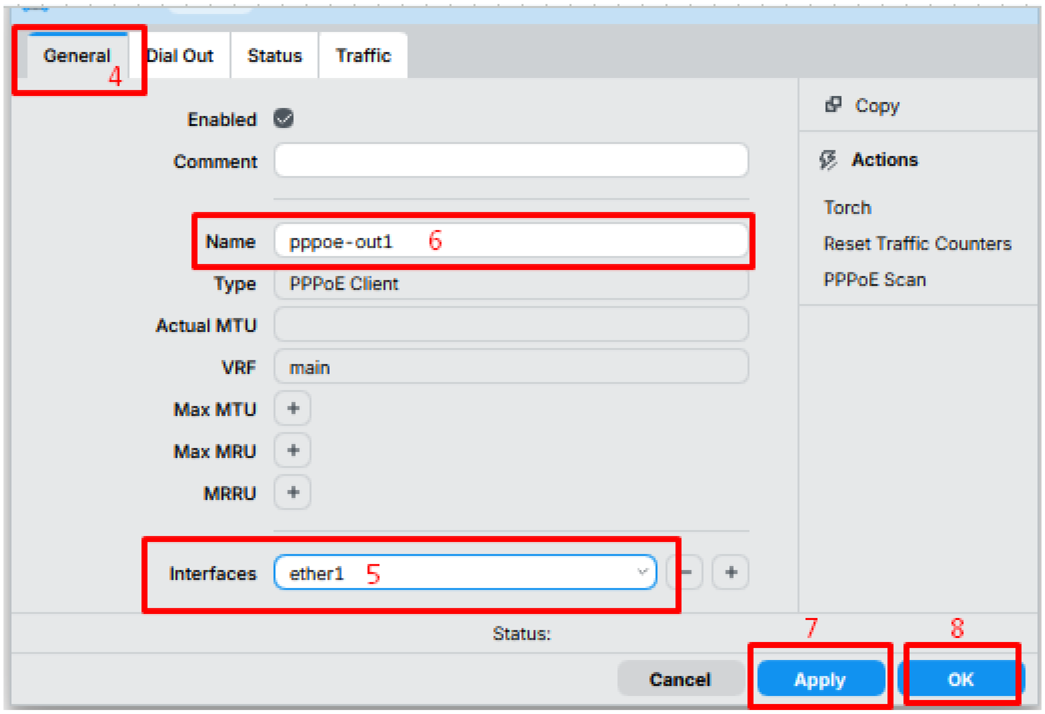
(Static IP: 27.147.239.52)

Winbox:

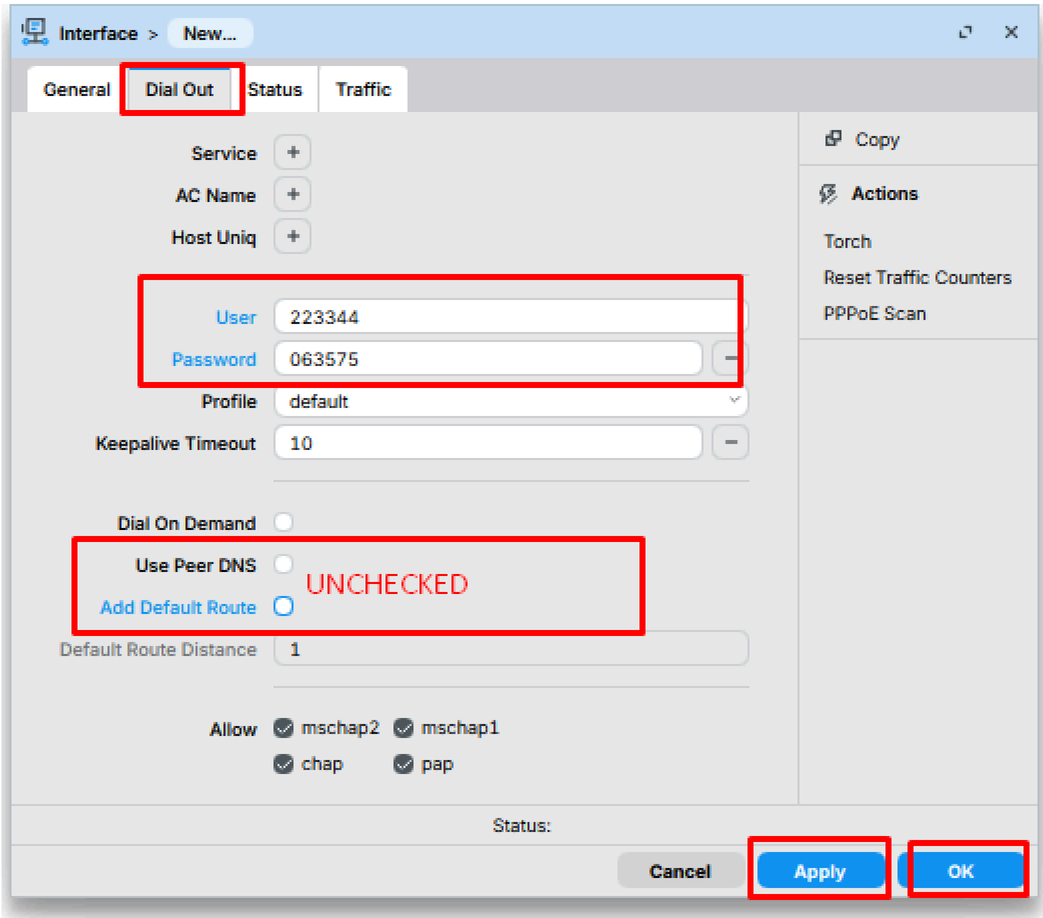
1. **Open Winbox → Go to Interfaces → PPP.**
2. **Click "+" → Select PPPoE Client.**



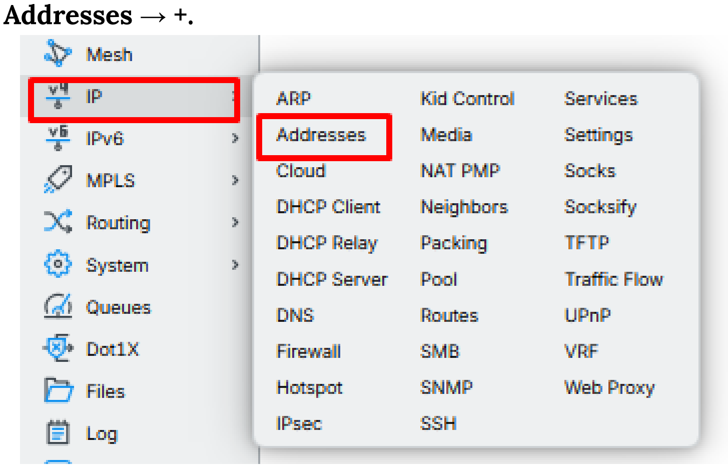
3. **In the General tab, choose your WAN interface (e.g., ether1).**



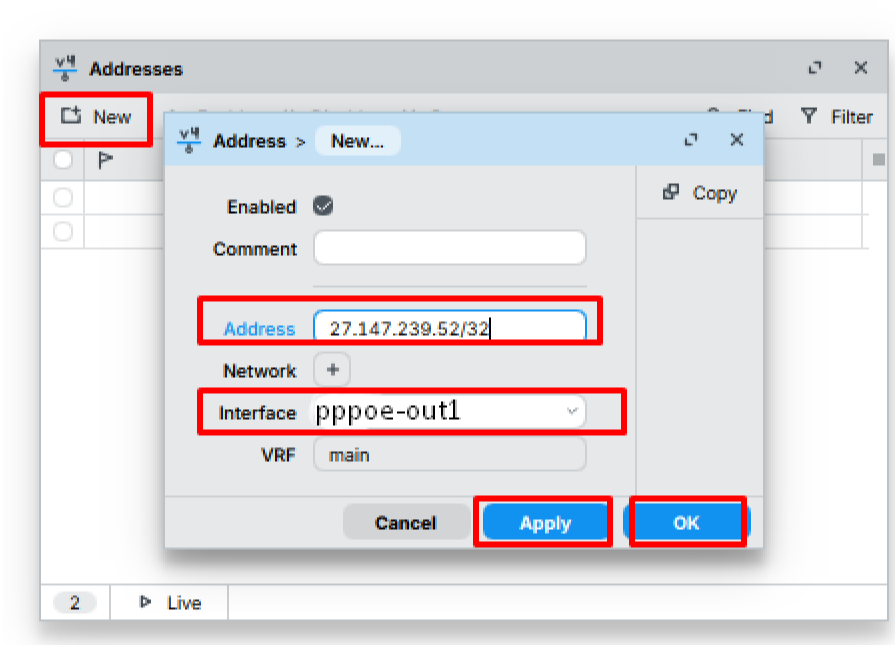
4. In Dial Out tab, enter:
    - Username = 223344
    - Password = 063575
    - Uncheck Add Default Route
    - Uncheck Use Peer DNS (optional)
5. Click Apply → OK.



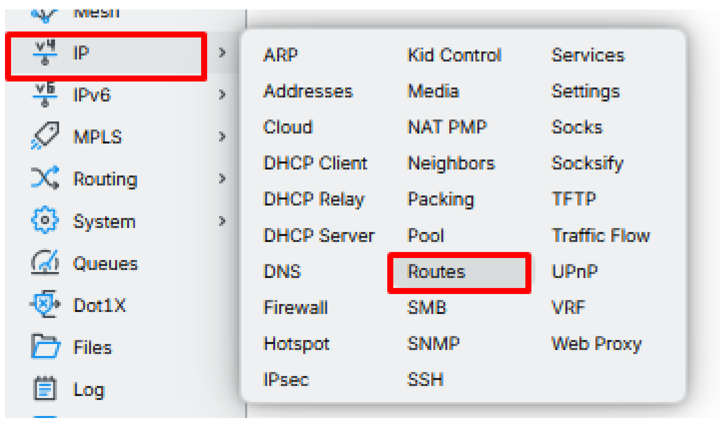
6. Go to IP → Addresses → +.



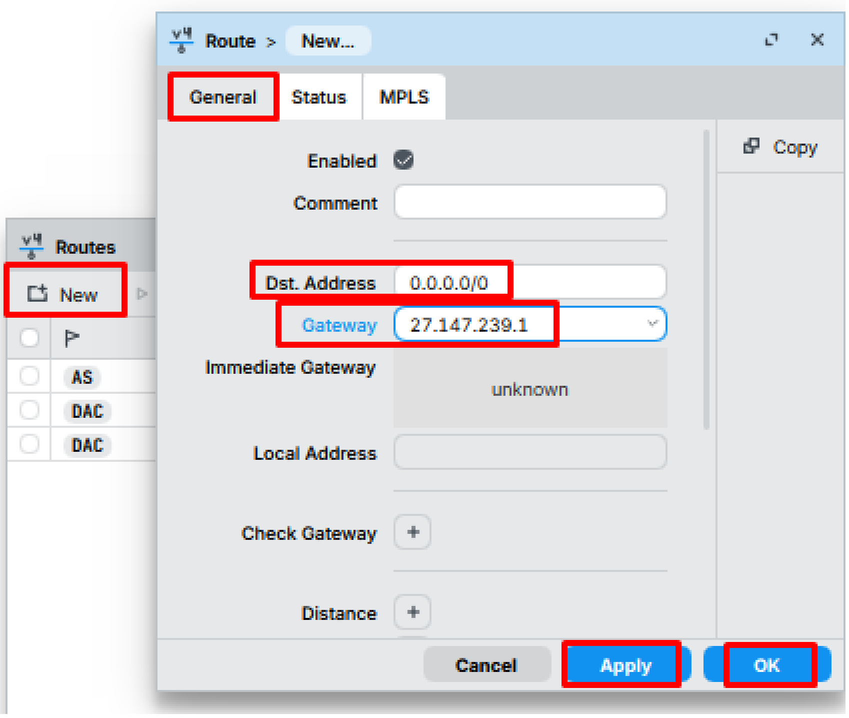
- Address = 27.147.239.52/32
- Interface = pppoe-out1
- Apply → OK.



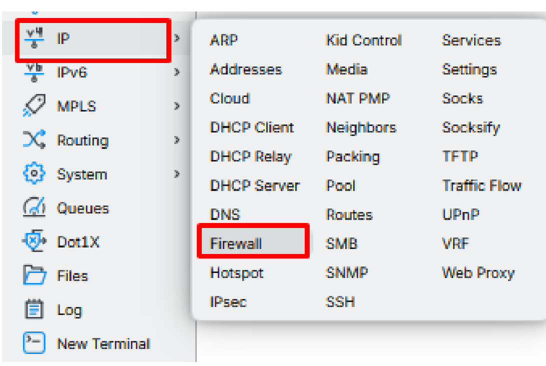
7. Go to IP → Routes → +.

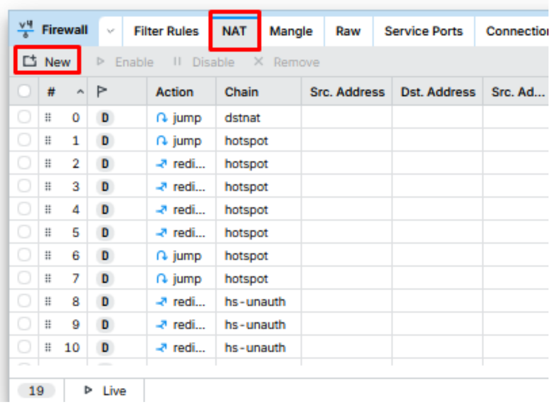


- Dst. Address = 0.0.0.0/0
- Gateway = ISP Provided Gateway (e.g., 27.147.239.1)
- Apply → OK.

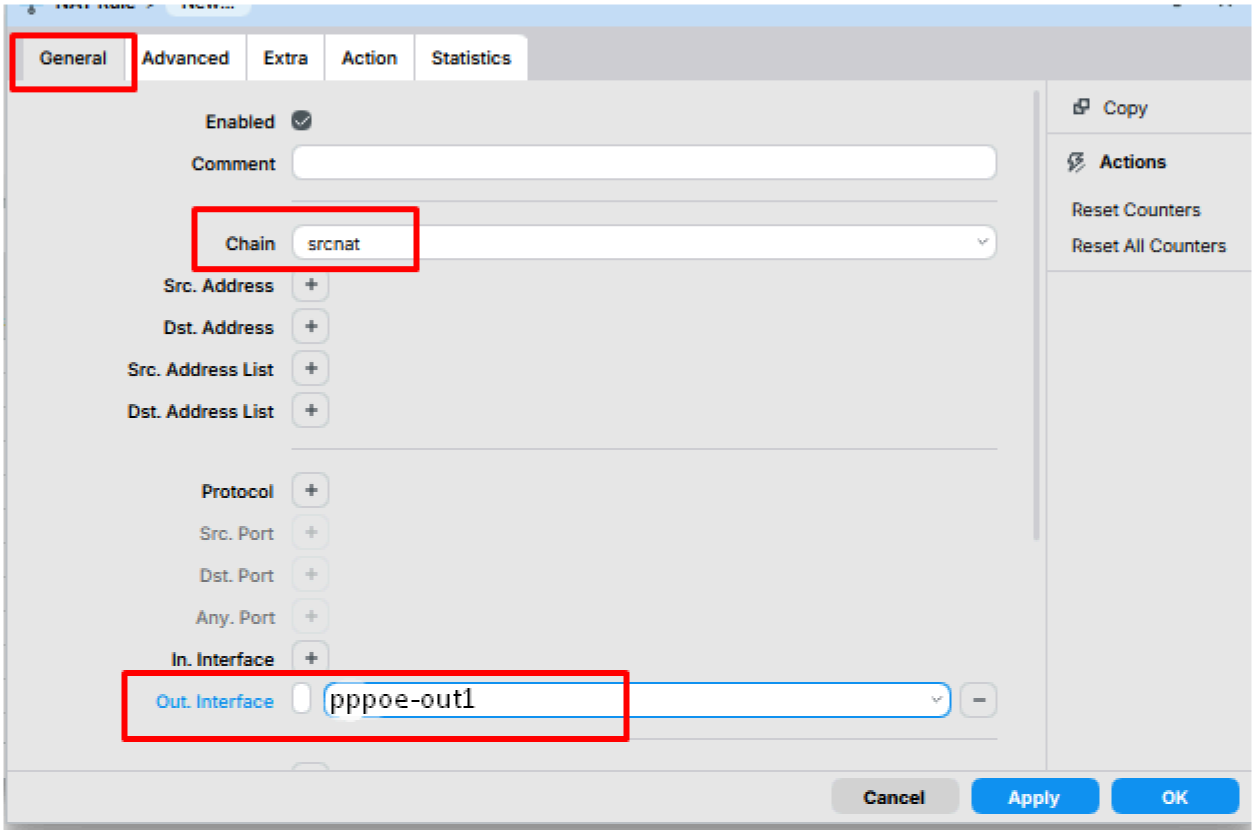


8. Go to IP → Firewall → NAT → +.

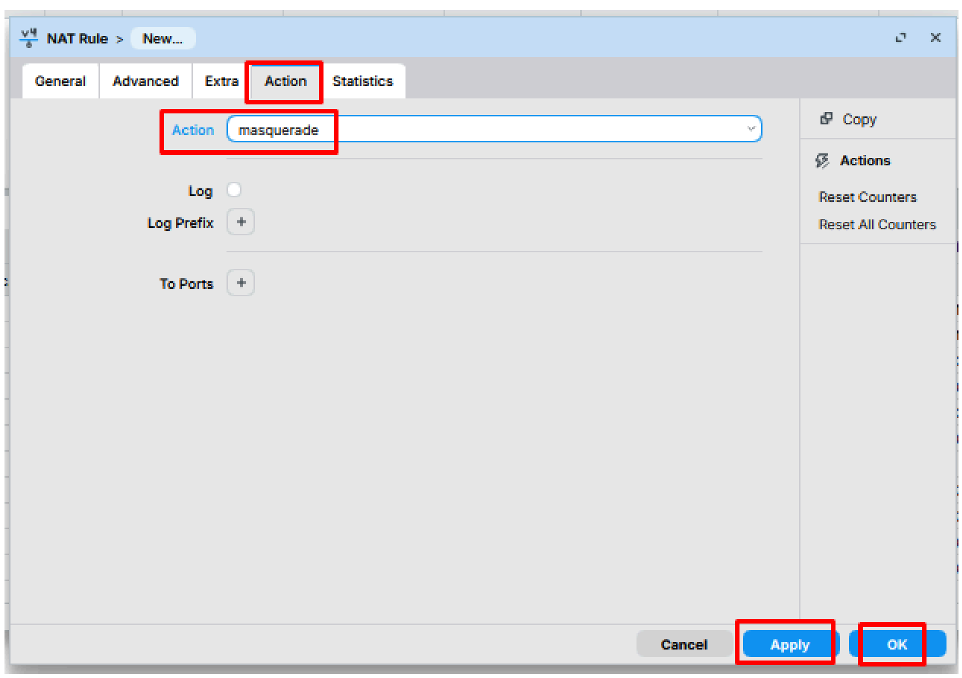




- Chain = srcnat
- Out. Interface = pppoe-out1



- **Action = masquerade**
- **Apply → OK.**



CLI:

```
/interface pppoe-client add name=pppoe-out1 interface=ether1 user=236403
  password=063575 use-peer-dns=no add-default-route=no disabled=no

/ip address add address=27.147.239.52/32 interface=pppoe-out1

/ip route add dst-address=0.0.0.0/0 gateway=27.147.239.1   # replace with your ISP
  gateway

/ip firewall nat add chain=srcnat out-interface=pppoe-out1 action=masquerade
```

Topic 3.3 – LAN Bridge + DHCP

Winbox:

1. Go to Bridge → Bridge → click "+".
   - Name = bridge-lan
   - Apply → OK.
2. Go to Bridge → Ports → click "+".
   - Interface = ether2, Bridge = bridge-lan.
   - Repeat for ether3 and ether4.
3. Go to IP → Addresses → click "+".
   - Address = 192.168.88.1/24
   - Interface = bridge-lan.
   - Apply → OK.
4. Go to IP → Pool → click "+".
   - Name = dhcp_pool
   - Ranges = 192.168.88.10-192.168.88.200
   - Apply → OK.
5. Go to IP → DHCP Server → click "DHCP Setup".
   - Select interface = bridge-lan
   - Next → Gateway = 192.168.88.1
   - Next → IP Pool = dhcp_pool
   - Next → Lease Time = 8h
   - Finish.

Now all devices on ether2, ether3, ether4 will get IPs and internet automatically.

CLI:

```
# Create bridge

/interface bridge add name=bridge-lan


# Add LAN ports to bridge

/interface bridge port add interface=ether2 bridge=bridge-lan

/interface bridge port add interface=ether3 bridge=bridge-lan

/interface bridge port add interface=ether4 bridge=bridge-lan


# Assign LAN IP to bridge

/ip address add address=192.168.88.1/24 interface=bridge-lan


# Create DHCP pool

/ip pool add name=dhcp_pool ranges=192.168.88.10-192.168.88.200


# DHCP server on bridge

/ip dhcp-server add name=dhcp1 interface=bridge-lan address-pool=dhcp_pool
lease-time=8h disabled=no


# DHCP network

/ip dhcp-server network add address=192.168.88.0/24 gateway=192.168.88.1
dns-server=8.8.8.8,8.8.4.4
```

✅ **Now your full setup is complete:**

- **WAN (ether1) = PPPoE (Dynamic or Static).**
- **LAN (ether2-4) = bridge with DHCP.**
- **NAT = masquerade on PPPoE interface.**
- **Gateway = automatic (Dynamic PPPoE) or manual (Static PPPoE).**

⚡ Note:

- For DHCP mode, the IP comes automatically from ISP.
- For Static mode, you must manually assign IP, Gateway, and DNS

Topic 3.4 – DNS Settings for LAN Clients

Winbox

1. Go to IP → DHCP Server → Networks.
2. Select the LAN network (e.g., 192.168.88.0/24) → Edit.
3. Set DNS Servers: 8.8.8.8, 8.8.4.4 (Google DNS) or your ISP DNS.
4. Apply → OK.

This ensures LAN clients receive proper DNS automatically.

CLI:

```
/ip dhcp-server network set 0 dns-server=8.8.8.8,8.8.4.4
```

Topic 3.5 – Firewall Basics

Purpose: Allow LAN → WAN traffic, block unwanted inbound traffic.

Winbox

1. Go to IP → Firewall → Filter Rules → +

2. Allow LAN → WAN:
   - Chain: forward
   - Src. Address: 192.168.88.0/24
   - Out. Interface: pppoe-out1
   - Action: accept

3. Drop everything else inbound from WAN:
   - Chain: forward
   - In. Interface: pppoe-out1
   - Action: drop

4. Apply → OK

   This creates a basic protective firewall for your network.

CLI:

```
/ip firewall filter add chain=forward src-address=192.168.88.0/24
  out-interface=pppoe-out1 action=accept

/ip firewall filter add chain=forward in-interface=pppoe-out1 action=drop
```

Topic 3.6 – Testing / Troubleshooting

1. Ping Internet:
   - Go to New Terminal → `ping 8.8.8.8`
   - If successful, WAN connection is working.
2. Check PPPoE Status:
   - Winbox: Interfaces → PPPoE Client → Status

   CLI:

   ```
   /interface pppoe-client print
   ```
3. Check DHCP Leases:
   - Winbox: IP → DHCP Server → Leases

   CLI:

   ```
   /ip dhcp-server lease print
   ```
4. Verify NAT:
   - Winbox: IP → Firewall → NAT

   CLI:

   ```
   /ip firewall nat print
   ```

Topic 3.7 – Optional: Wi-Fi / Access Point (AP) Settings

If ether2–4 connect to an AP or MikroTik wireless interface:

Winbox

1. Go to Wireless → Interfaces → wlan1 → Enable
2. Set Mode: AP Bridge
3. Set SSID: e.g., MyLAN_WiFi
4. Security Profile → Set WPA2 Pre-Shared Key
5. Bridge WLAN to bridge-lan if needed (IP → Bridge → Ports → Add wlan1)

CLI

```
/interface wireless set wlan1 mode=ap-bridge ssid=MyLAN_WiFi disabled=no

/interface bridge port add interface=wlan1 bridge=bridge-lan

/interface wireless security-profiles set [ find default=yes ]
  wpa2-pre-shared-key="YourPassword"
```

Heading 4: Firewall and Hotspot Configuration

Topic 4.1 – Basic Firewall Protection (Drop Invalid & Bogus Traffic)

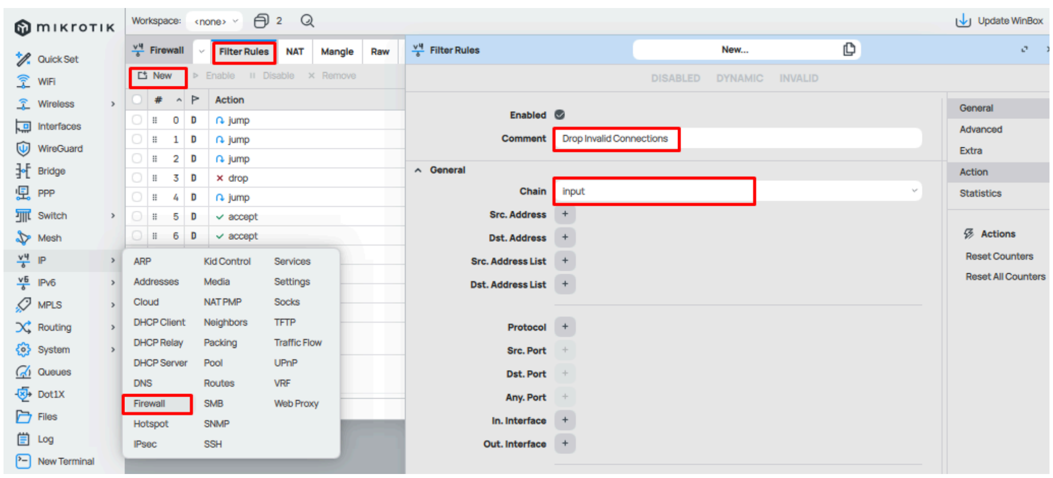
Step 1: Drop Invalid Packets

CLI:

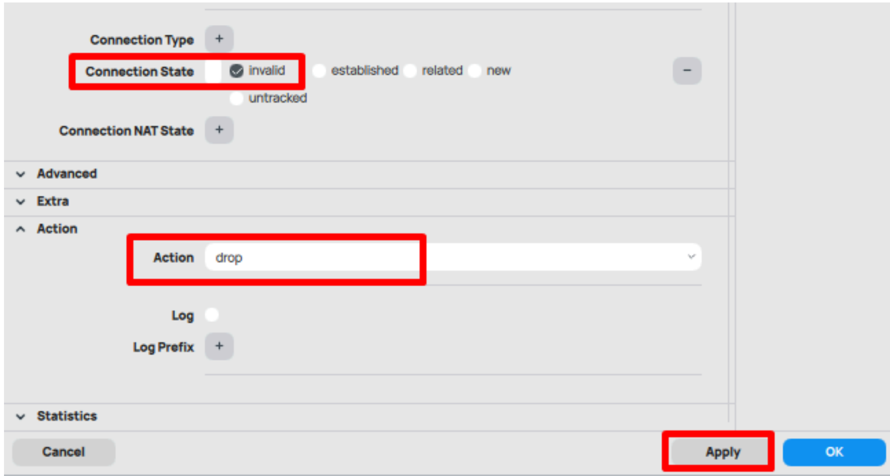
/ip firewall filter add chain=input connection-state=invalid action=drop comment="Drop invalid connections"

Winbox:

- Go to IP → Firewall → Filter Rules → Add (+)
- Chain: input



- Connection State: invalid
- Action: drop
- Comment: Drop invalid connections



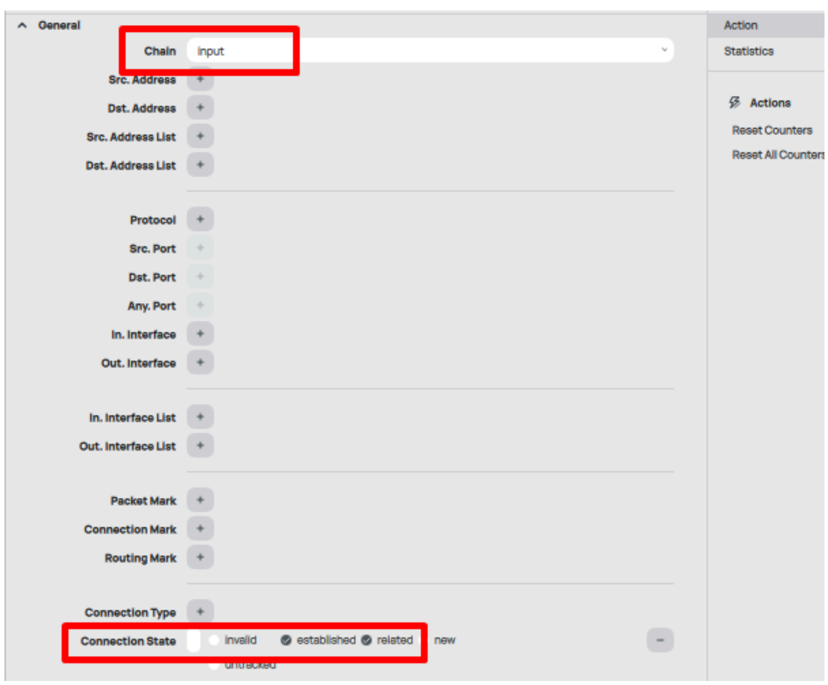
Step 2: Allow Established & Related Connections

CLI:

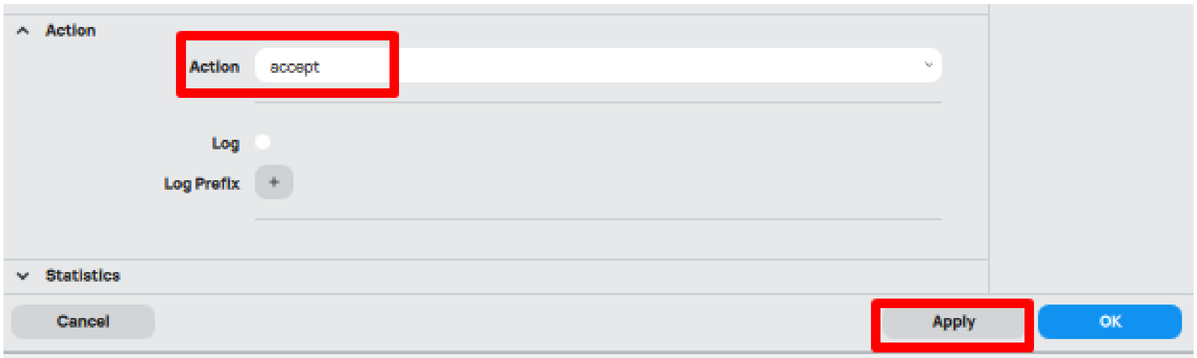
```
/ip firewall filter add chain=input connection-state=established,related action=accept comment="Allow Established/Related"
```

Winbox:

- Chain: input
- Connection State: established,related



- **Action:** accept
- **Comment:** Allow Established/Related



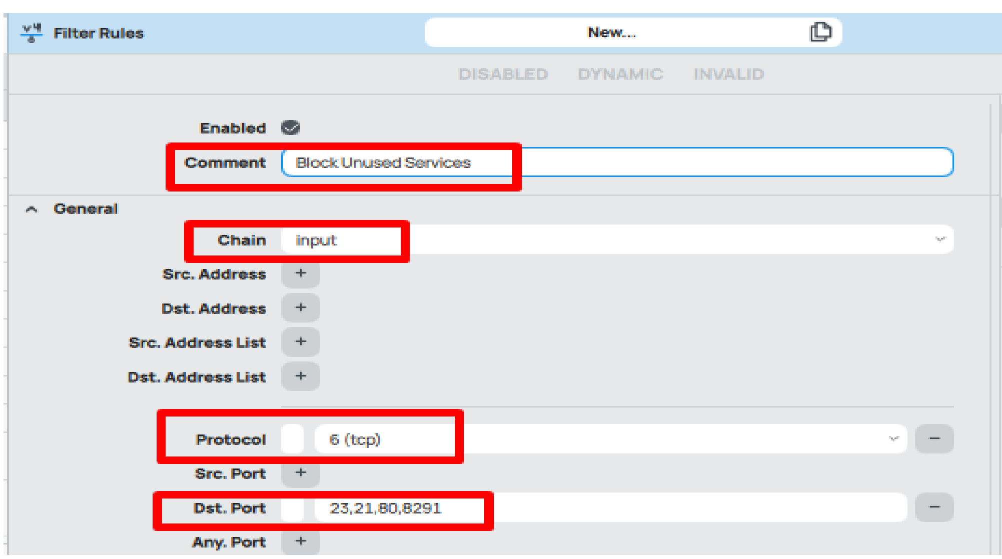
Step 3: Drop Bogus Traffic to Router

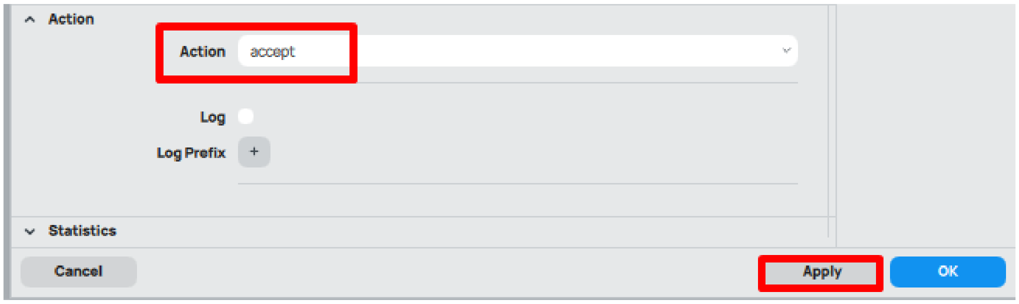
CLI:

```
/ip firewall filter add chain=input protocol=tcp dst-port=23,21,80,8291 action=drop
comment="Block Unused Services"
/ip service disable telnet
/ip service disable ftp
/ip service disable www
```

Winbox:

- **IP → Firewall → Filter Rules → Add (+)**
  Protocol: TCP, Dst-Port: 23,21,80,8291, Action: drop, **Comment:** Block Unused Services





- IP → Services → Disable: Telnet, FTP, WWW

Topic 4.2 – Block Ping Flood (ICMP Protection)

CLI:

```
/ip firewall filter add chain=input protocol=icmp limit=50/5s,2 action=accept
    comment="Allow limited ping"
/ip firewall filter add chain=input protocol=icmp action=drop comment="Drop Excessive
    Ping"
```

Winbox:

- Rule 1: Accept ICMP with limit (50 per 5s)
- Rule 2: Drop all other ICMP

Topic 4.3 – Protect Router from Port Scans

CLI:

```
/ip firewall filter add chain=input protocol=tcp psd=21,3s,3,1 action=drop comment="Drop
        Port Scans"
/ip firewall filter add chain=input protocol=tcp tcp-flags=fin,!syn,!rst,!psh,!ack,!urg
        action=drop comment="Drop TCP FIN Scans”
```

Winbox:

- IP → Firewall → Filter Rules → Add (+)
- Use PSD for port scan
- Drop FIN scans

Topic 4.4 – Secure Remote Access (Winbox/SSH only from Your IP)

CLI:

```
/ip firewall filter add chain=input protocol=tcp dst-port=22,8291 src-address=Your.Public.IP
        action=accept comment="Allow Admin IP Only"
```

```
/ip firewall filter add chain=input protocol=tcp dst-port=22,8291 action=drop
    comment="Drop Other SSH/Winbox"
```

Winbox:

- Add rule: Allow SSH (22) & Winbox (8291) only from your public IP
- Add rule: Drop all other SSH/Winbox

Topic 4.5 – Hotspot Configuration (with WiFi SSID Redirect)

Step 1: Create Hotspot on WiFi/LAN Interface

CLI:

```
/ip hotspot setup

    # Follow prompts:

    # Select interface (e.g., LAN_Bridge)
    # Local address (e.g., 192.168.88.1/24)
    # Address pool (e.g., 192.168.88.2-192.168.88.254)
    # Select certificate (none if not using SSL)
    # IP of SMTP server (leave default)
    # DNS servers (e.g., 8.8.8.8)
    # DNS name (e.g., hotspot.local)
    # Create admin user (e.g., admin / password)
```

Winbox:

- Go to IP → Hotspot → Hotspot Setup
- Interface: select WiFi SSID interface LAN_Bridge)
- Address pool: e.g., 192.168.88.0/24
- DNS servers: (ISP DNS or 8.8.8.8, 1.1.1.1)
- DNS name: hotspot.local
- Finish → MikroTik will auto-create DHCP + NAT + Firewall rules.

Step 2: Hotspot User Creation

CLI:

```
/ip hotspot user add name=user1 password=1234 profile=default
```

Winbox:

- Go to IP → Hotspot → Users → Add (+)
- Username: user1
- Password: 1234

Step 3: Set Idle Timeout

CLI:

```
/ip hotspot profile set [find default=yes] idle-timeout=30m
```

Winbox:

- Go to IP → Hotspot → Servers → Profiles
- Double click on the default profile
- Find Idle Timeout → set to 30m
- Click OK

Step 4: WiFi SSID → Hotspot Login Redirect

When a device connects to the configured WiFi SSID (bound to Hotspot interface):

- Device connects → requests any website (e.g., google.com).
- MikroTik intercepts → redirects to http://hotspot.local/login.
- The user enters credentials.
- Router authenticates → internet access is granted.

Topic 4.6 – Bonus Topic to Understand the Chain Rules(Firewall Basics)

A chain is like a *processing path* or *set of rules* that traffic passes through in the firewall. When a packet (data) arrives, MikroTik decides which chain should handle it, based on where the packet is going.

- Input chain → Traffic going to the router itself
  (Example: you connect with Winbox, ping the router, or SSH into it).

- Forward chain → Traffic passing through the router
  (Example: your PC in LAN browsing the internet via the router).

- Output chain → Traffic generated by the router itself
  (Example: the router updates time with NTP, sends DNS queries, or pings another server).

👉 In short:

- Input = To router
- Forward = Through router
- Output = From router

Heading 5: PPPoE Server Setup for ISPs

Scenario:

VLAN	Gateway/Local IP	IP Pool
501	10.192.50.1	10.192.50.10-200
502	10.152.29.1	10.192.29.10-200
503	10.192.248.1	10.192.248.10-200

Physical port: ether2 connected to subscriber switch

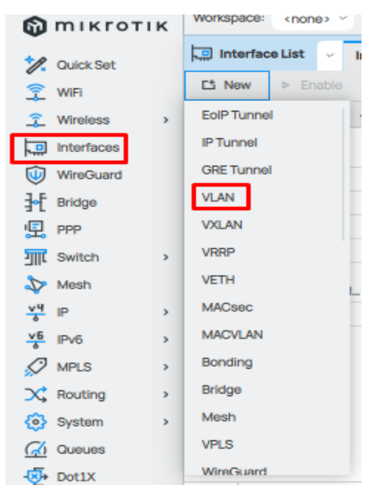
Topic 5.1 - VLAN Interfaces

CLI

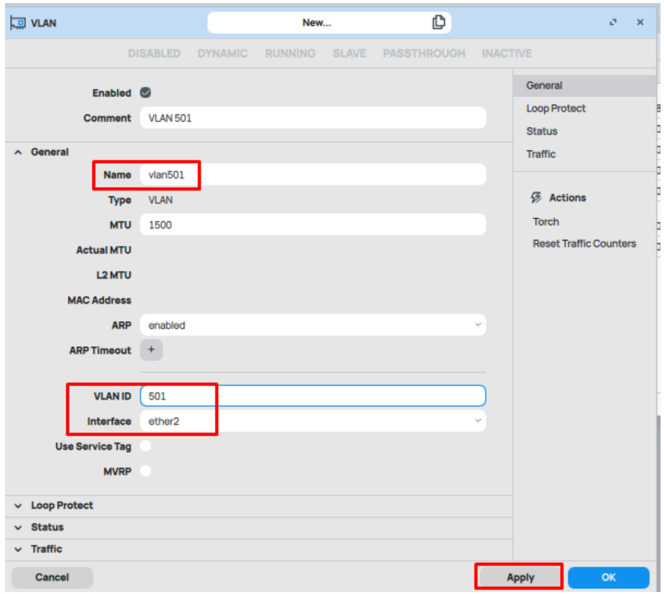
```
/interface vlan
    add name=vlan501 interface=ether2 vlan-id=501
    add name=vlan502 interface=ether2 vlan-id=502
    add name=vlan503 interface=ether2 vlan-id=503
/ip address(Gateways)
    add address=10.192.50.1/24 interface=vlan501
    add address=10.152.29.1/24 interface=vlan502
    add address=10.192.248.1/24 interface=vlan503
```

Winbox

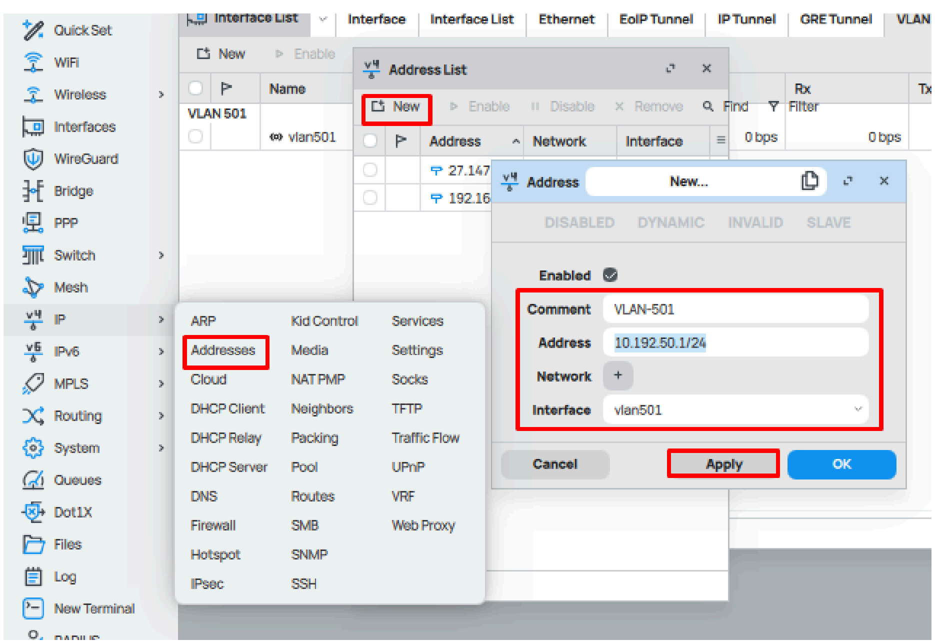
- **Interfaces → VLAN → +**



- Name: vlan501
- Interface: ether2
- VLAN ID: 501 → OK



- **Repeat for VLAN 502 and 503**
- **Assign IP: IP → Addresses → + → Select VLAN interface → Enter gateway → OK**



Topic 5.2 - IP Pools

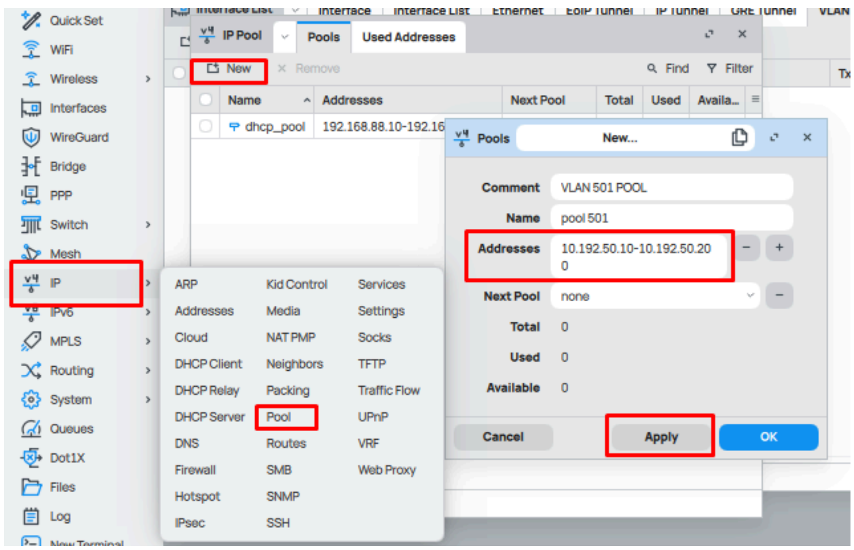
CLI

/ip pool

add name=pool501 ranges=10.192.50.10-10.192.50.200
add name=pool502 ranges=10.152.29.10-10.152.29.200
add name=pool503 ranges=10.192.248.10-10.192.248.200

Winbox

- IP → Pool → + → Name: pool501 → Range: 10.192.50.10-200 → OK



- Repeat for pool502 and pool503

Topic 5.3 - PPP Profiles

CLI

/ppp profile

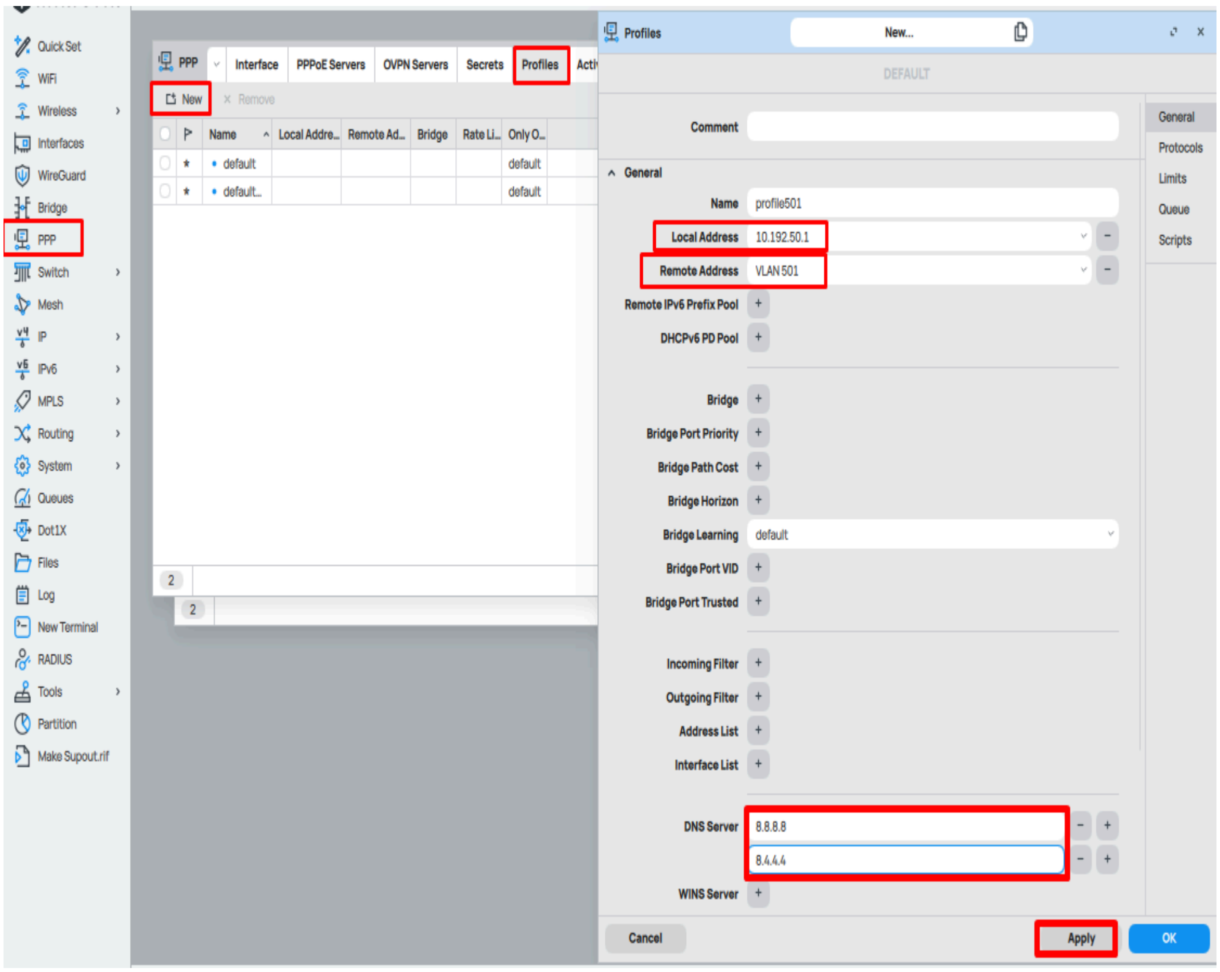
add name=profile501 local-address=10.192.50.1 remote-address=pool501 dns-server=8.8.8.8,1.1.1.1

add name=profile502 local-address=10.152.29.1 remote-address=pool502 dns-server=8.8.8.8,1.1.1.1

add name=profile503 local-address=10.192.248.1 remote-address=pool503 dns-server=8.8.8.8,1.1.1.1

Winbox

- **PPP → Profiles → New(+) → Name: profile501 → Local Address: 10.192.50.1 → Remote Pool: pool501 → DNS: 8.8.8.8,1.1.1.1 → OK**



- **Repeat for profile502 and profile503**

Topic 5.4 - One PPP Secret

CLI

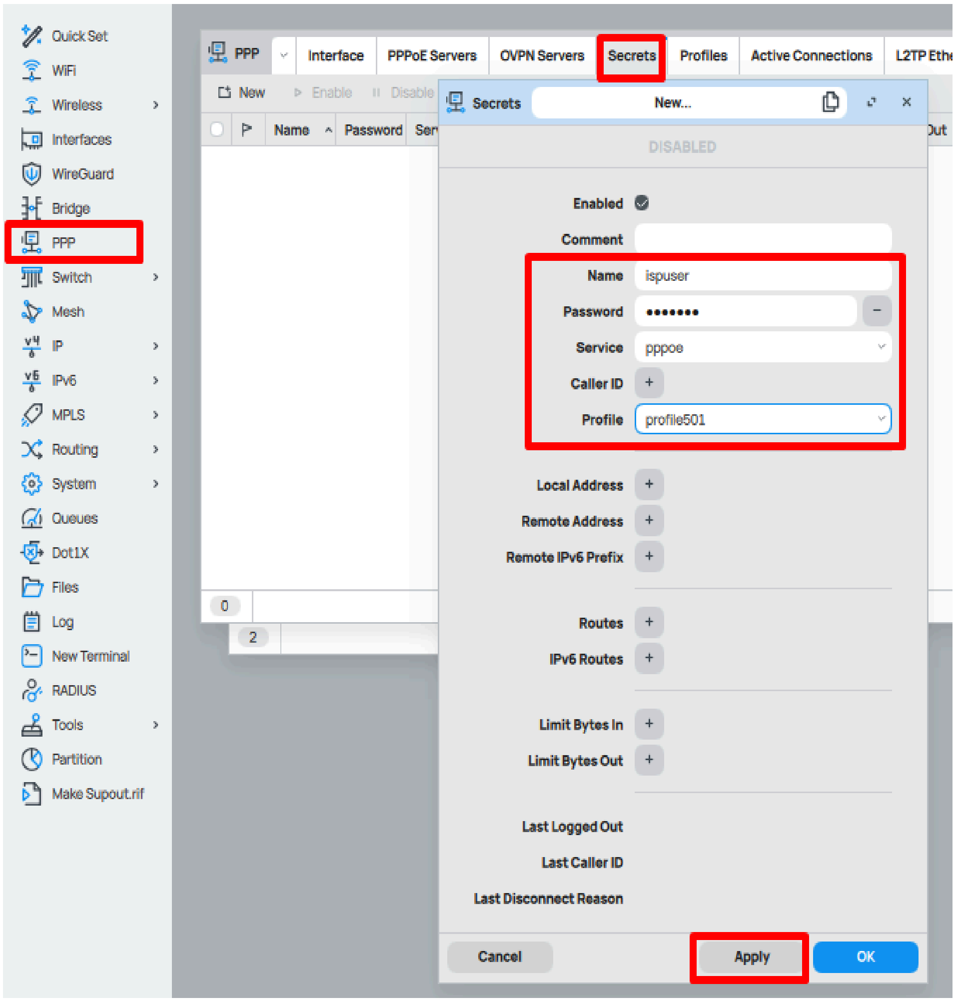
/ppp secret

add name=ispuser password=isppass service=pppoe profile=default

Winbox

- **PPP → Secrets → +**
  Name: ispuser
  Password: isppass

Service: PPPoE
Profile: profile501 → OK



Note: This secret will use profile501. For VLAN502 or 503, you can either duplicate the secret or adjust the profile dynamically (advanced setup)

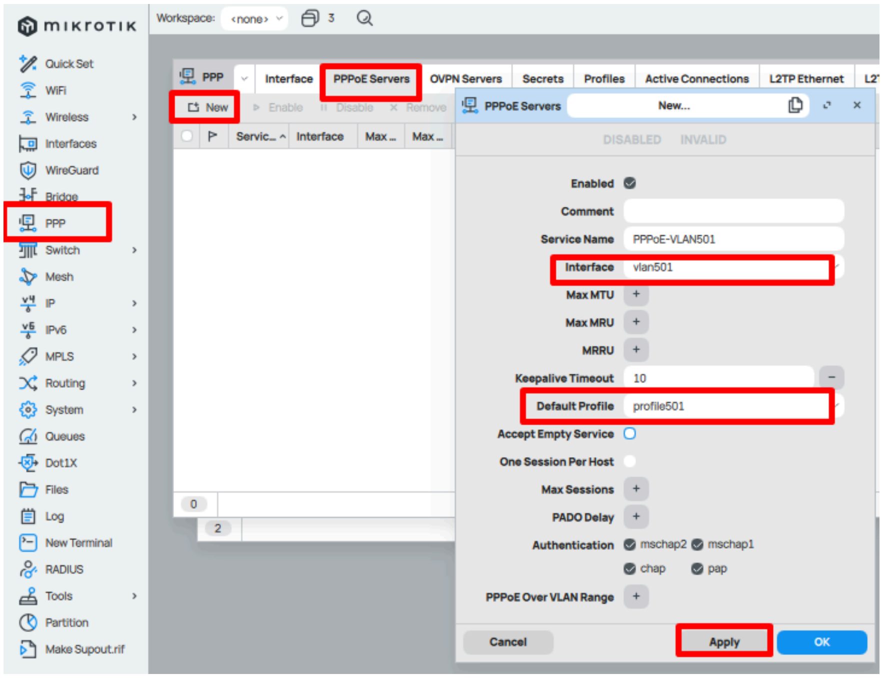
Topic 5.5 - PPPoE Servers

CLI

```
/interface pppoe-server server
    add service-name=PPPoE-VLAN501 interface=vlan501 default-profile=profile501 disabled=no
    add service-name=PPPoE-VLAN502 interface=vlan502 default-profile=profile502 disabled=no
    add service-name=PPPoE-VLAN503 interface=vlan503 default-profile=profile503 disabled=no
```

Winbox

- **PPP → Interface → + → PPPoE Server**
  Interface: vlan501
  Service Name: PPPoE-VLAN501
  Default Profile: profile501 → Enable → OK



- **Repeat for VLAN502 and VLAN503**

Topic 5.6 - NAT / Internet Access

CLI

/ip firewall nat
add chain=srcnat out-interface=ether1 action=masquerade

Winbox

- **IP → Firewall → NAT → +**
  - **Chain: srcnat**
  - **Out Interface: ether1 (WAN)**
  - **Action: masquerade → OK**

Topic 5.7 - Verify PPP Connections

CLI

```
/ppp active print
```

Winbox

- **PPP → Active**

  Shows: username, virtual PPP interface, IP assigned, uptime, traffic

Topic -5.8 - Optional: Bandwidth Limiting

CLI

```
/queue simple add name=ispuser target=pppoe-out1 max-limit=2M/2M
```

Winbox

- **Queues → Simple Queues → + → Target: select PPPoE interface → Max Limit → OK**

✅ **Result**

- **VLAN501 → PPPoE server → pool 10.192.50.x → gateway 10.192.50.1**
- **VLAN502 → PPPoE server → pool 10.152.29.x → gateway 10.152.29.1**
- **VLAN503 → PPPoE server → pool 10.192.248.x → gateway 10.192.248.1**
- **Single PPP secret ispuser can log in (via VLAN501 in this setup)**
- **Client receives IP, dynamic PPP interface is created, and Internet works via NAT**

Heading 6: Bandwidth Management

Topic 6.1 – Simple Queue (Per User / Per IP Limit)

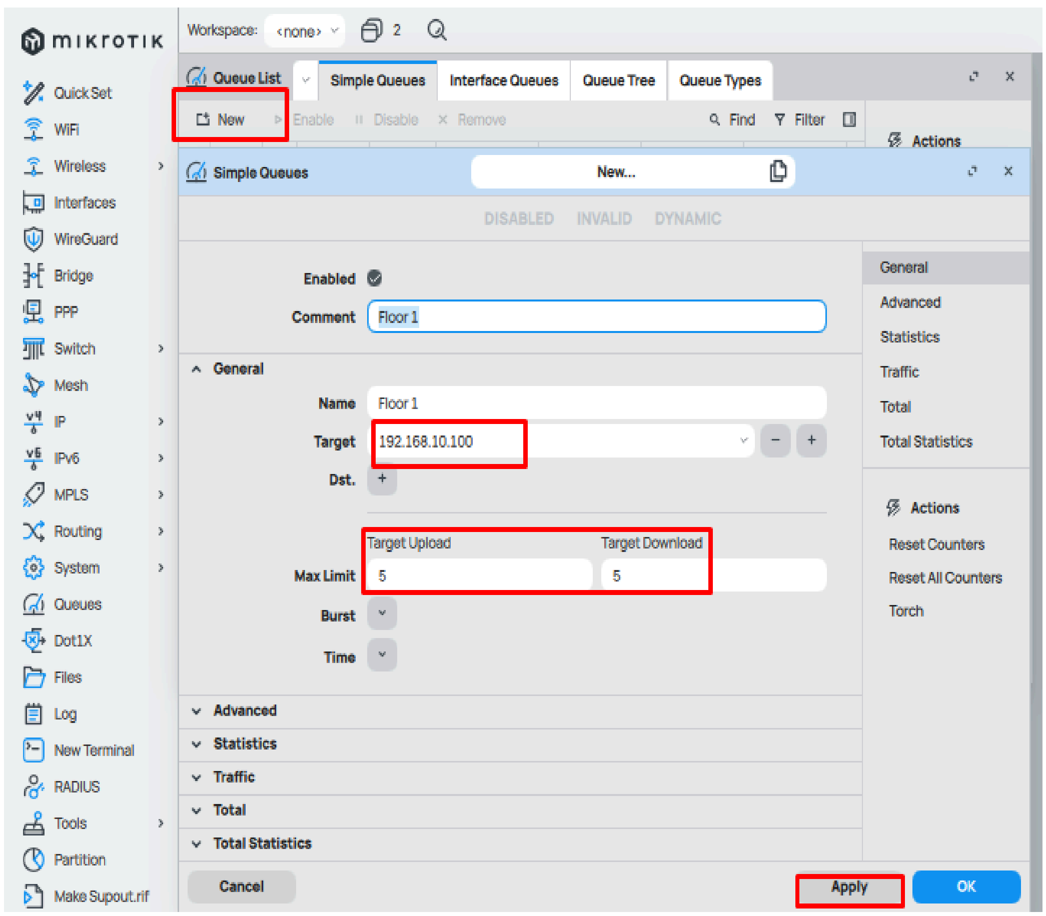
Example: Limit a single IP

CLI:

```
/queue simple add name=User1 target=192.168.10.100/32 max-limit=2M/2M
```

Winbox:

Go to Queues → Simple Queues → Add (+)
Name: Floor1
Target: 192.168.10.100
Max Limit: 5M/5M



✅ **This ensures that 192.168.10.100 cannot exceed 5 Mbps Upload / 5 Mbps Download.**

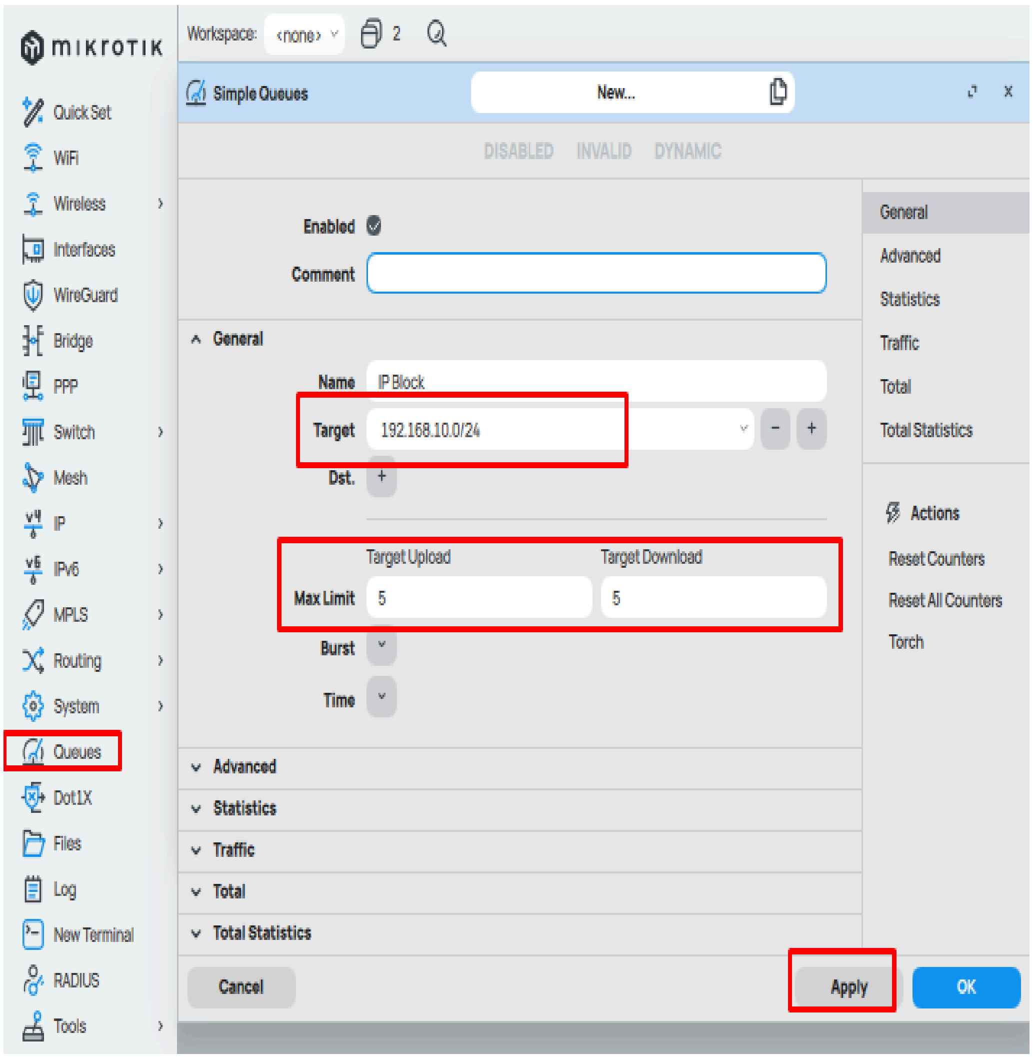
Example: Limit whole subnet

CLI:

```
/queue simple add name=LAN-Limit target=192.168.10.0/24 max-limit=10M/10M
```

Winbox:

Target: 192.168.10.0/24
Max Limit: 5M/5M



Topic 6.2 – Queue Tree (Advanced Bandwidth Control)

Queue Tree works with parent–child structure and requires Mangle rules.

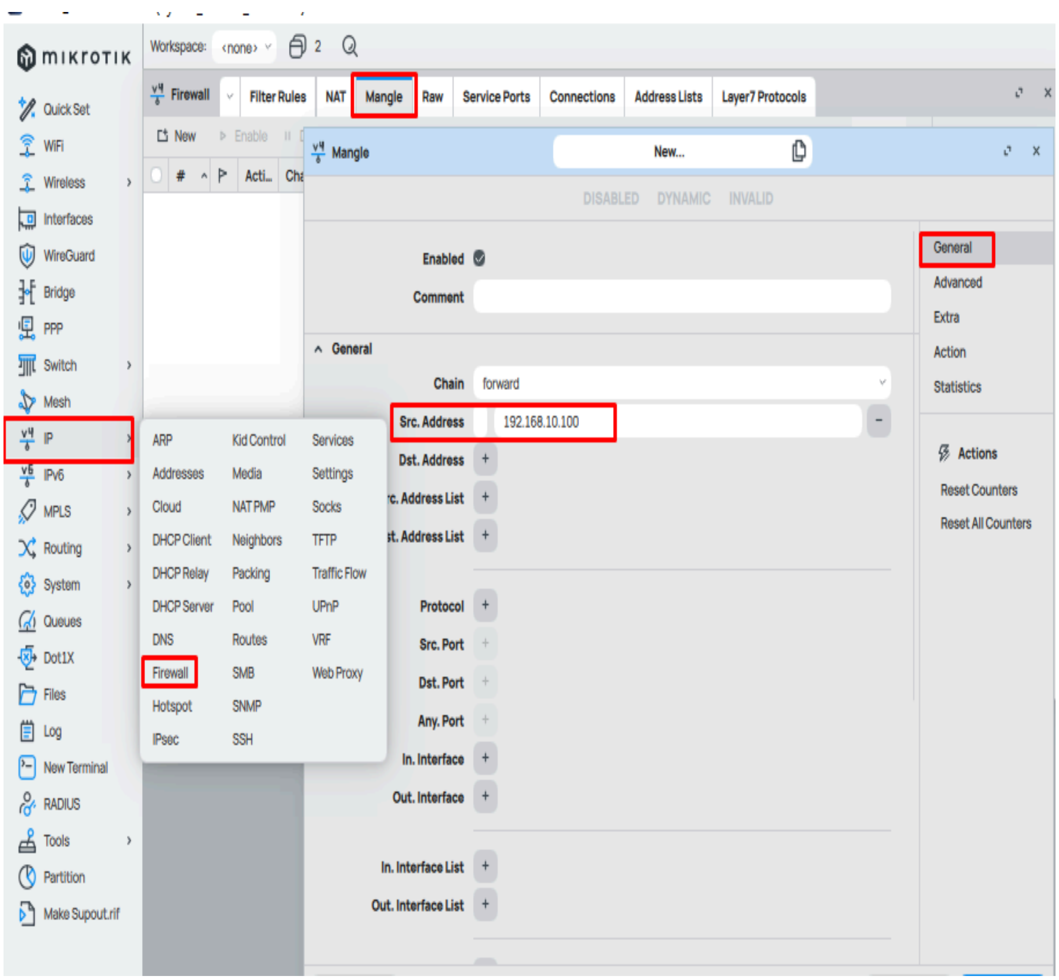
Step 1: Mark Packets by Source Address

CLI:

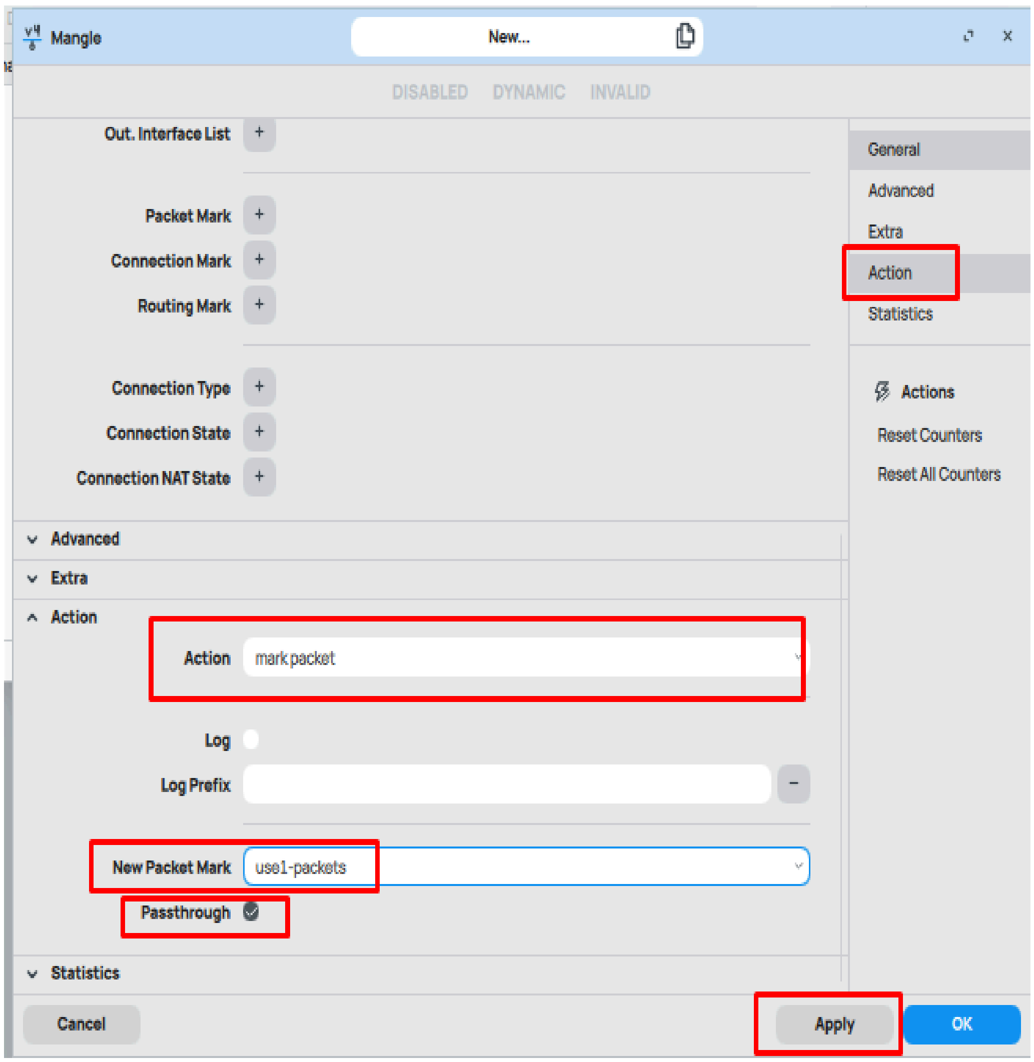
```
/ip firewall mangle add chain=forward src-address=192.168.10.100 action=mark-packet
 new-packet-mark=User1-Packets passthrough=yes
```

Winbox:

- **Open Winbox → go to IP → Firewall → Mangle.**
- **Click + (Add New Rule).**
- **In the General tab:**
  - **Chain = forward**
  - **Src. Address = 192.168.10.100**



- Go to the Action tab:
    - Action = mark-packet
    - New Packet Mark = User1-Packets (type the name)
    - Tick Passthrough = yes
    - Click OK.



✅ Now all packets from 192.168.10.100 are marked as User1-Packets.

Step 2: Create Queue Tree for User

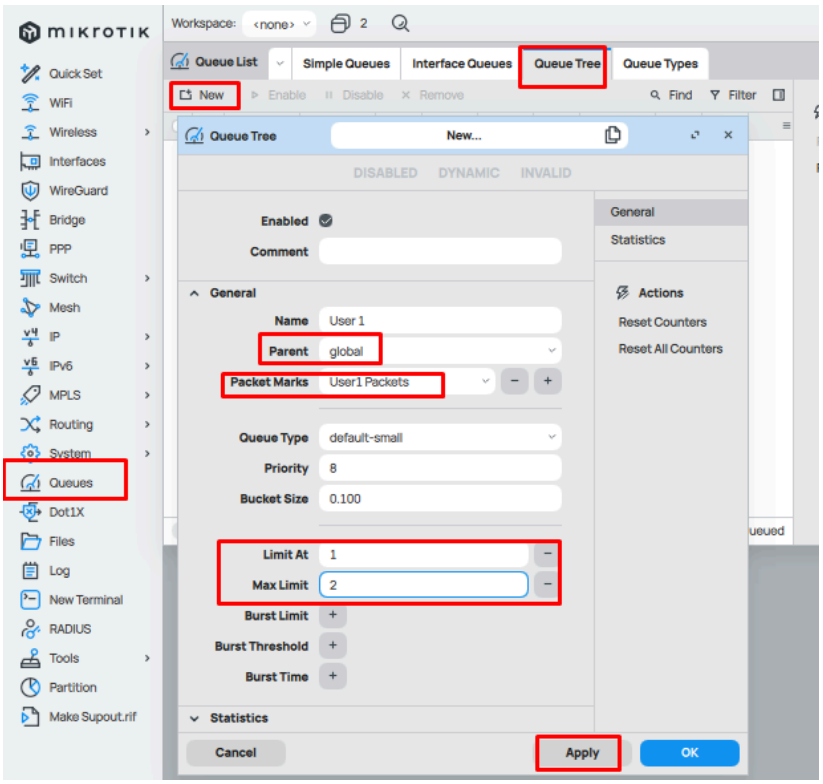
CLI:

```
/queue tree add name=User1 parent=global packet-mark=User1-Packets limit-at=1M max-limit=2M
```

Winbox:

- In Winbox → go to Queues → Queue Tree.

- Click + (Add New Queue Tree).
- Fill in the details:
    - Name = User1
    - Parent = global (or global-total depending on your version)
    - Packet Mark = User1-Packets (select from dropdown)
    - Limit At = 1M
    - Max Limit = 2M
- Click OK.



✅ Now the queue will control the traffic for that IP based on the packet mark.

Topic 6.3 – Burst Limit (Smart Bandwidth Sharing)

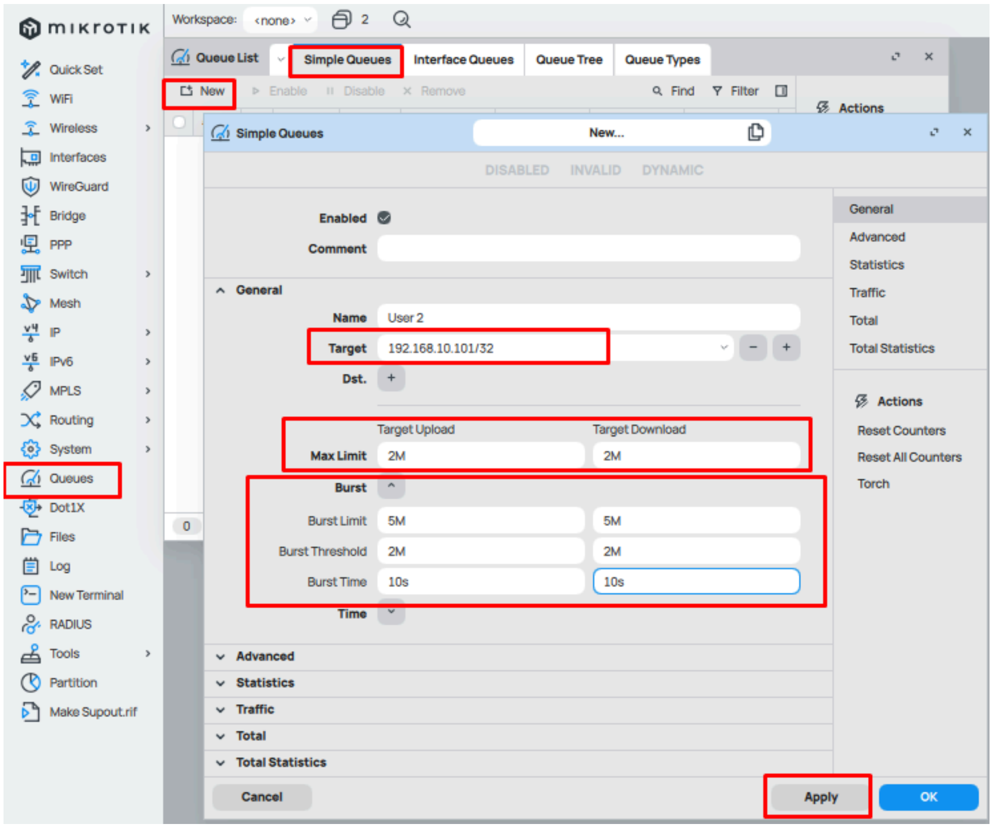
Example: Give users 2 Mbps normally, but allow burst up to 5 Mbps for first 10 seconds

CLI:

/queue simple add name=User2 target=192.168.10.101/32 max-limit=2M/2M burst-limit=5M/5M burst-threshold=2M/2M burst-time=10s/10s

Winbox:

Max Limit: 2M/2M
Burst Limit: 5M/5M
Burst Threshold: 2M/2M
Burst Time: 10s/10s



✅ **User enjoys 5 Mbps for 10s when idle, then drops back to 2 Mbps.**

Topic 6.4 – Priority Bandwidth (QoS for Important Traffic)

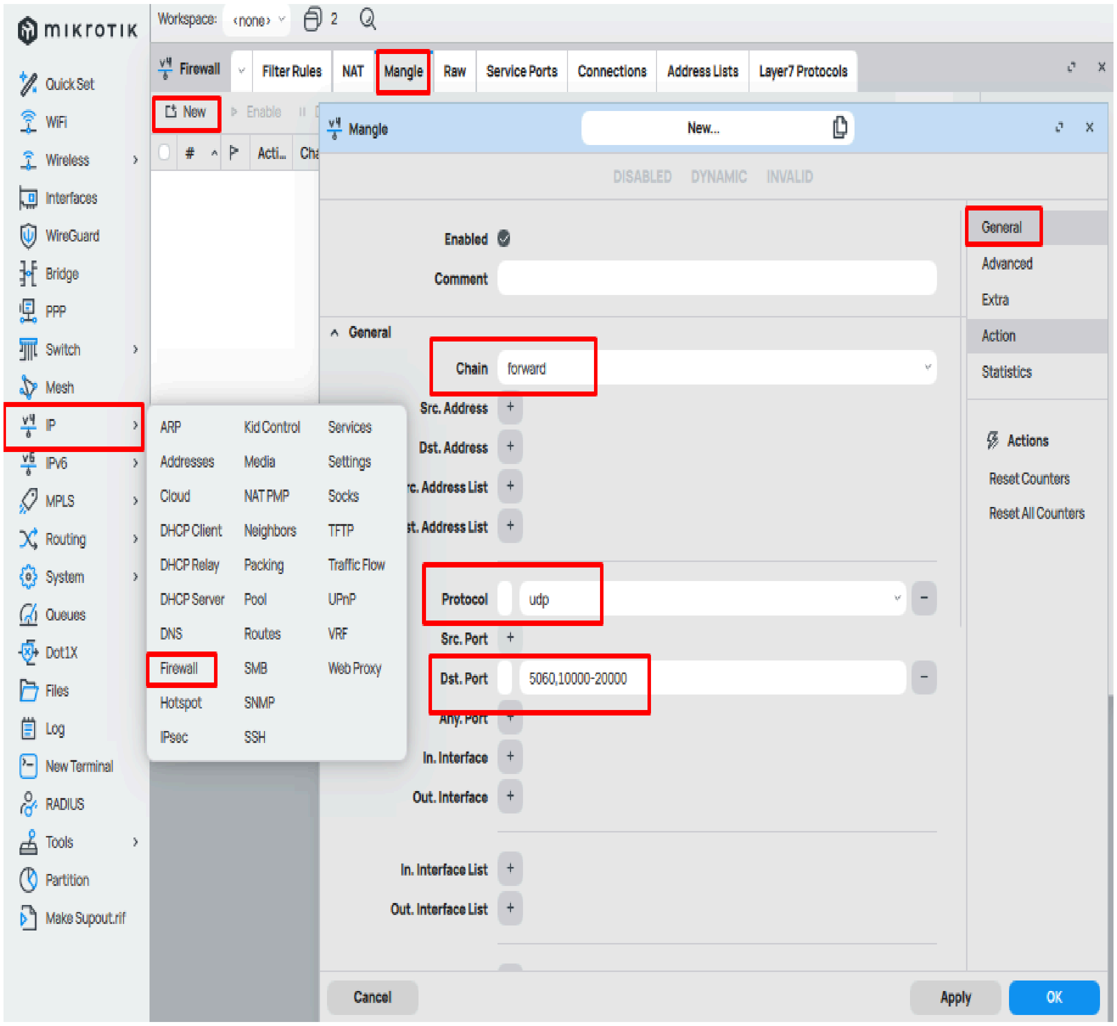
Step 1: Mark Packets (e.g., VoIP priority)

CLI:

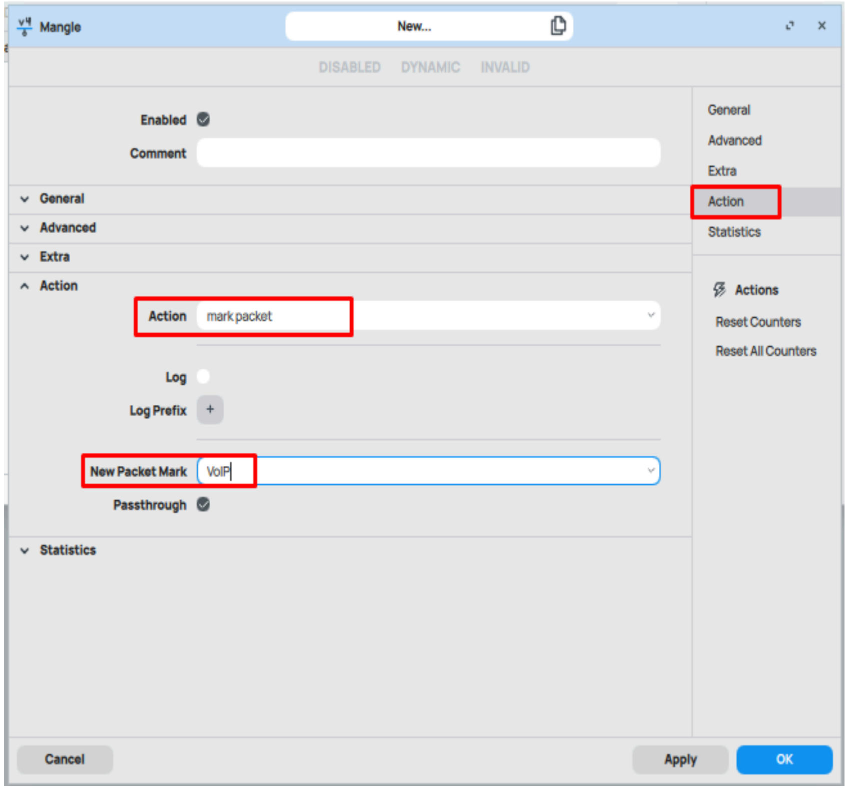
```
/ip firewall mangle add chain=forward protocol=udp port=5060,10000-20000
action=mark-packet new-packet-mark=VoIP passthrough=yes
```

Winbox:

- Go to IP → Firewall → Mangle.
- Click ➕ (Add New).
- In the General tab:
  - Chain = forward
  - Protocol = udp
  - Dst. Port = 5060,10000-20000



- **In the Action tab:**
  - **Action = mark-packet**
  - **New Packet Mark = type VoIP**
  - **Passthrough = ✔️ (checked)**
- **Click OK.**



Step 2: Create Queue Tree with Priority

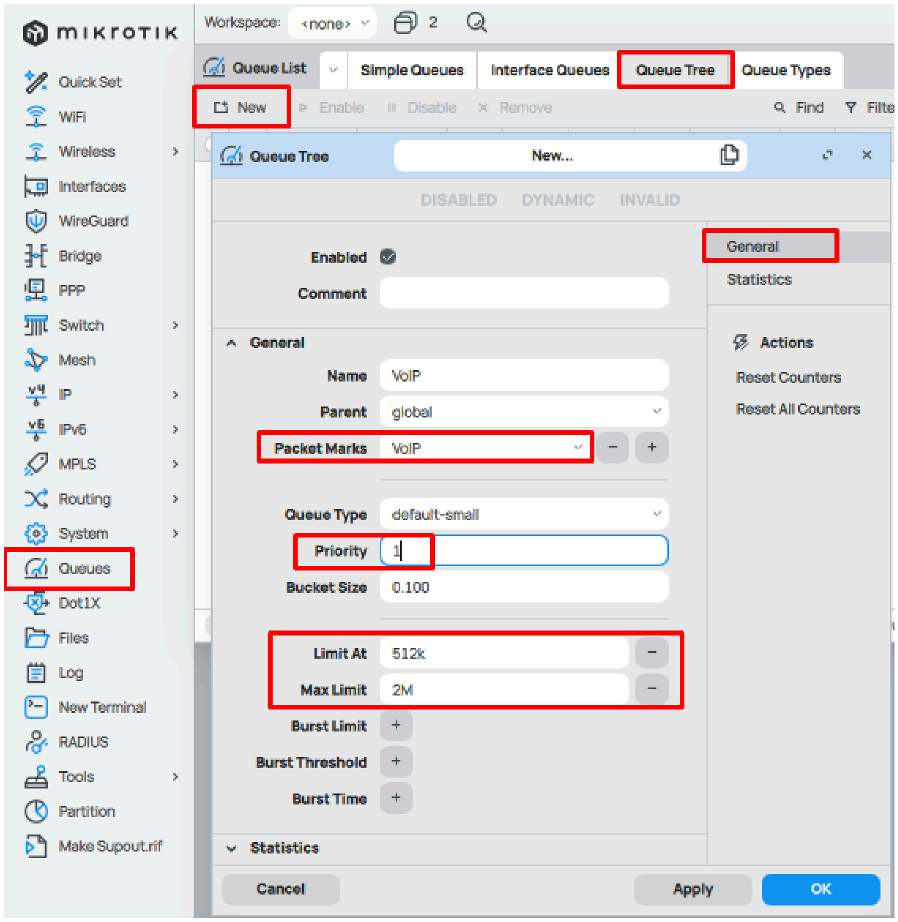
CLI:

/queue tree add name=VoIP parent=global packet-mark=VoIP limit-at=512k max-limit=2M priority=1

Winibox:

Go to Queues → Queue Tree.

- **Click ➕ (Add New).**
- **In the General tab:**
  - **Name = VoIP**
  - **Parent = global (or global-total in ROS v7)**
  - **Packet Mark = select VoIP**
  - **Limit At = 512k**
  - **Max Limit = 2M**
  - **Priority = 1**
- **Click OK.**



✅ **VoIP traffic gets highest priority, while others share leftover bandwidth.**

Topic 6.5 – Example: Bandwidth Plan for ISP Users

- Profile 1Mbps → 1M/1M
- Profile 2Mbps → 2M/2M
- Profile 5Mbps → 5M/5M

CLI:

```
/queue simple add name=User1 target=192.168.10.101/32 max-limit=1M/1M
/queue simple add name=User2 target=192.168.10.102/32 max-limit=2M/2M
/queue simple add name=User3 target=192.168.10.103/32 max-limit=5M/5M
```

Winbox:

User1 (192.168.10.101 → 1M/1M)

- Open Winbox → Queues → Simple Queues.
- Click ➕ (Add New).
- In the General tab:
  - Name = User1
  - Target = 192.168.10.101/32

- **In the Advanced tab (optional) → leave default.**
  In the Queue tab → leave default.
- **In the Max Limit fields:**
    - **Target Upload Max Limit = 1M**
    - **Target Download Max Limit = 1M**
- **Click OK.**

✅ **This method is commonly used by small ISPs for customer bandwidth control.**

Heading 7: MikroTik Hotspot Complete Setup(ISP LEVEL)

Topic 7.1 – Assign LAN/Bridge IP for Hotspot

Hotspot চালানোর জন্য LAN interface-এ একটি gateway IP দরকার।

Winbox:

IP → Addresses → Add (+)

- **Address: 192.168.50.1/24**
- **Interface: bridge1**
- **Apply → OK**

CLI:

/ip address add address=192.168.50.1/24 interface=bridge1

Ensure bridge1-এ LAN ports/WiFi add করা আছে

Topic 7.2 – Create IP Pool

Winbox:

Go to IP → Pool → Add (+)

- **Name: Hotspot-Pool**
- **Ranges: 192.168.50.10-192.168.50.254**
  Click Apply → OK

CLI:

Hotspot user-দের IP দেওয়ার জন্য pool তৈরি করতে হবে।

/ip pool add name=Hotspot-Pool ranges=192.168.50.10-192.168.50.254

Topic 7.3 - WAN Configuration (Choose ONE)

Topic 7.3A – Static Public IP

Winbox:

Go to IP → Addresses → Add (+)

- **Address: 10.250.46.230/30**
- **Interface: ether1**

Go to IP → Routes → Add (+)

- Gateway: 10.250.46.229

CLI:

ISP থেকে fixed IP পেলে:

```
/ip address add address=10.250.46.230/30 interface=ether1
/ip route add gateway=10.250.46.229
```

Topic 7.3B – Private IP (DHCP)

Winbox:

Go to IP → DHCP Client → Add (+)

- Interface: ether1
- ✔ Add Default Route
- ✖ Use Peer DNS (optional off)

CLI:

Upstream router থেকে IP নিলে:

```
/ip dhcp-client add interface=ether1 add-default-route=yes use-peer-dns=no
```

Topic 7.3C – PPPoE WAN

Winbox:

Go to PPP → Interface → PPPoE Client → Add (+)

- Interface: ether1
- Username: 236403
- Password: 123456
- ✔ Add Default Route
- ✖ Use Peer DNS

CLI:

ISP username/password দিয়ে:

```
/interface pppoe-client add name=pppoe-out1 interface=ether1 user=236403 password=123456 add-default-route=yes use-peer-dns=no disabled=no
```

Topic 7.4 – Setup Hotspot Server

Winbox:

Go to IP → Hotspot → Hotspot Setup

Follow:

- **Interface:** bridge1
- **Address Pool:** Hotspot-Pool
- **Certificate:** none
- **DNS Name:** hotspot.local
- **User:** admin / 12345

Click Next → Finish

CLI:

```
/ip hotspot setup
```

Follow prompts:

```
interface: bridge1
local address: 192.168.50.1/24
address pool: 192.168.50.10-192.168.50.254
select certificate: none
dns servers: 8.8.8.8
dns name: hotspot.local
user: admin
password: 12345
```

Topic 7.5 – Configure DNS

Winbox:

Go to IP → DNS

- **Servers:** 8.8.8.8, 1.1.1.1
- **✔ Allow Remote Requests**

Click Apply → OK

CLI:

```
/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes
```

Topic 7.6 – Configure NAT

Winbox:

Go to IP → Firewall → NAT → Add (+)

For Static/Private:

- **Chain:** **srcnat**
- **Out Interface:** **ether1**
- **Action:** **masquerade**

For PPPoE:

- **Out Interface:** **pppoe-out1**

CLI:

Static / Private IP:

```
/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
```

PPPoE:

```
/ip firewall nat add chain=srcnat out-interface=pppoe-out1 action=masquerade
```

Topic 7.7 – Create Hotspot Users

Winbox:

Go to IP → Hotspot → Users → Add (+)

- **Name:** **admin**
- **Password:** 12345

Repeat for user1

CLI:

```
/ip hotspot user add name=admin password=12345
/ip hotspot user add name=user1 password=123
```

Topic 7.8 – Admin + User System

Winbox:

Go to System → Users → Add (+)

- Name: admin
- Password: 12345
- Group: full

CLI:

Router admin (management access):

```
/user add name=admin password=12345 group=full
```

Admin can:

- Winbox login
- User control
- Monitoring

Topic 7.9 – Monitor Users

Winbox:

Go to:

- IP → Hotspot → Active (online users)
- IP → Hotspot → Hosts (connected devices)

CLI:

Active users:

```
/ip hotspot active print
```

Detailed:

```
/ip hotspot active print detail
```

Connected devices:

```
/ip hotspot host print
```

Disconnect user:

```
/ip hotspot active remove [find user=user1]
```

Disable user:

```
/ip hotspot user disable user1
```

Topic 7.10 – Auto Disconnect Idle Users

Winbox:

IP → Hotspot → User Profiles → Default

- **Idle Timeout:** **1h**

CLI:

/ip hotspot user profile set default idle-timeout=1h

১ ঘন্টা ব্যবহার না করলে **auto logout**

Topic 7.11 – Troubleshooting

❌ **Login page** আসছে না:

/ip hotspot enable hotspot1

❌ **Internet** নাই:

/ip firewall nat print

❌ **DNS problem:**

/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes

❌ **PPPoE down:**

/interface pppoe-client enable pppoe-out1

❌ **IP** পাচ্ছে না:

/ip pool print

Common Mistakes (Explain)	
Problem	**Result**
NAT wrong interface	**Internet** কাজ করবে না
DNS missing	**Login page** আসবে না
Bridge not configured	**User connect** করবে না
PPPoE down	**Internet** আসবে না

Topic 7.12 – Summary

Checklist:

/ip address print
/ip route print
/ip dns print
/ip firewall nat print
/ip hotspot print

Static vs Private vs PPPoE (Clear Concept)

Static vs Private vs PPPoE (Clear Concept)		
Type	Meaning	Command
Static IP	ISP fixed IP দেয়	/ip address
Private IP	অন্য router থেকে IP	/ip dhcp-client
PPPoE	Username/password login	/interface pppoe-client

Final Result:

- User WiFi connect করবে
- Redirect → hotspot.local
- Login → Internet
- Admin সব monitor করতে পারবে
- Idle user auto disconnect

Summary of this Chapter:

✔ Full ISP Hotspot System
✔ Admin Control
✔ User Monitoring
✔ Auto Timeout System

Heading 8: Multi-WAN Failover (5-Port MikroTik Complete Guide)

Topic 8.1 - Chapter Objective

After, finalizing the chapter, we can do-

- ২টা **ISP** দিয়ে **failover** করতে
- **PPPoE / Static / DHCP / Private** সব বুঝতে
- **Winbox + CLI** দুইভাবে **setup** করতে
- **real ISP troubleshooting** করতে

Topic 8.2 - Failover Concept

Failover মানে:

Primary ISP down হলে **Backup ISP automatically** চালু হবে

Router কীভাবে **decide** করে?
distance=1 → Primary, distance=2 → Backup

Router সবসময় কম **distance route use** করে

Topic 8.3 - Network Design (5 Port MikroTik)

Port Plan	
Port	**Use**
ether1	**ISP1 (Primary)**
ether2	**ISP2 (Backup)**
ether3	**LAN**
ether4	**LAN**
ether5	**LAN**

Topic 8.4 - LAN Setup

Winbox

1. **Bridge → Add → bridge-lan**
2. **Bridge → Ports → Add → ether3,4,5**
3. **IP → Address → Add → 192.168.88.1/24**

CLI

```
/interface bridge add name=bridge-lan
/interface bridge port add bridge=bridge-lan interface=ether3
/interface bridge port add bridge=bridge-lan interface=ether4
/interface bridge port add bridge=bridge-lan interface=ether5
/ip address add address=192.168.88.1/24 interface=bridge-lan
```

Topic 8.5 - Common Rules (সব scenario-র জন্য)

Rule 1: NAT must

Winbox

IP → Firewall → NAT → Add

- **Chain: srcnat**
- **Out Interface: WAN**
- **Action: masquerade**

CLI

```
/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
/ip firewall nat add chain=srcnat out-interface=ether2 action=masquerade
```

Rule 2: Auto Route OFF

- **PPPoE → Add Default Route** ❌
- **DHCP → Add Default Route** ❌

কারণ: আমরা **manual control** করবো

Rule 3: Distance

Primary = 1, Backup = 2

Topic 8.6 - Scenario 1: PPPoE + PPPoE

Winbox

1. **PPPoE Client → ether1 + ether2**
2. **Add Default Route OFF**
3. **NAT → দুইটা**
4. **Route:**
    - **pppoe-out1 → distance=1**
    - **pppoe-out2 → distance=2**

CLI

```
/interface pppoe-client add name=pppoe-out1 interface=ether1 user=u1 password=p1 add-default-route=no

/interface pppoe-client add name=pppoe-out2 interface=ether2 user=u2 password=p2 add-default-route=no
```

```
/ip firewall nat add chain=srcnat out-interface=pppoe-out1 action=masquerade

/ip firewall nat add chain=srcnat out-interface=pppoe-out2 action=masquerade
```

```
/ip route add gateway=pppoe-out1 distance=1

/ip route add gateway=pppoe-out2 distance=2
```

Topic 8.7 - Scenario 2: PPPoE + DHCP

Winbox

1. **PPPoE → ether1**
2. **DHCP → ether2**
3. **Add Default Route OFF**
4. **NAT → both**
5. **Route:**
    - **pppoe-out1 → 1**
    - **ether2 → 2**

CLI

```
/interface pppoe-client add name=pppoe-out1 interface=ether1 user=u password=p
add-default-route=no

/ip dhcp-client add interface=ether2 add-default-route=no


/ip firewall nat add chain=srcnat out-interface=pppoe-out1 action=masquerade

/ip firewall nat add chain=srcnat out-interface=ether2 action=masquerade


/ip route add gateway=pppoe-out1 distance=1

/ip route add gateway=ether2 distance=2
```

Topic 8.8 - Scenario 3: PPPoE + Static Failover

Winbox

Step 1: PPPoE (Primary)

- **Interfaces → PPPoE Client → +**
  - **Interface:** ether1
  - **Username / Password**
  - ❌ **Uncheck: Add Default Route**

Step 2: Static IP (Backup)

- **IP → Addresses → +**
  - **Address:** 203.76.120.10/30
  - **Interface:** ether2

Step 3: NAT

- **IP → Firewall → NAT → +**

Rule 1:

- **Out Interface:** pppoe-out1
- **Action:** masquerade

Rule 2:

- **Out Interface:** ether2
- **Action:** masquerade

Step 4: Route

- **IP → Routes → +**

Primary:

- **Gateway:** pppoe-out1
- **Distance:** 1

Backup:

- **Gateway:** 203.76.120.9
- **Distance:** 2

CLI

```
/interface pppoe-client add name=pppoe-out1 interface=ether1 user=u password=p
add-default-route=no

/ip address add address=203.76.120.10/30 interface=ether2

/ip firewall nat add chain=srcnat out-interface=pppoe-out1 action=masquerade
/ip firewall nat add chain=srcnat out-interface=ether2 action=masquerade

/ip route add gateway=pppoe-out1 distance=1 comment="Primary"
/ip route add gateway=203.76.120.9 distance=2 comment="Backup"
```

Topic 8.9 - Scenario 4: PPPoE + Private

Winbox

Step 1: PPPoE

Step 1: PPPoE (Primary)

- **Interfaces → PPPoE Client → +**
  - **Interface:** ether1

- Username / Password
- ❌ Uncheck: Add Default Route

Step 2: Private IP

- IP → Addresses → +
  - Address: 192.168.1.2/24
  - Interface: ether2

Step 3: NAT

- IP → Firewall → NAT → +

Rule 1:

- Out Interface: pppoe-out1
- Action: masquerade

Rule 2:

- Out Interface: ether2
- Action: masquerade

Step 4: Route

Primary:

- Gateway: pppoe-out1
- Distance: 1

Backup:

- Gateway: 192.168.1.1
- Distance: 2

CLI

```
/interface pppoe-client add name=pppoe-out1 interface=ether1 user=u password=p
add-default-route=no

/ip address add address=192.168.1.2/24 interface=ether2

/ip firewall nat add chain=srcnat out-interface=pppoe-out1 action=masquerade
/ip firewall nat add chain=srcnat out-interface=ether2 action=masquerade

/ip route add gateway=pppoe-out1 distance=1
```

```
/ip route add gateway=192.168.1.1 distance=2
```

Topic 8.10 - Scenario 5: Static + Dynamic (DHCP)

Winbox

Step 1: Static IP(Primary)

- **IP → Addresses → +**
  - **Address:** 203.76.120.10/30
  - **Interface:** ether1

Step 2: DHCP Client

- **IP → DHCP Client → +**
  - **Interface:** ether2
  - ❌ **Uncheck: Add Default Route**

Step 3: NAT

- **ether1 → masquerade**
- **ether2 → masquerade**

Step 4: Route

Primary:

- **Gateway:** 203.76.120.9
- **Distance:** 1

Backup:

- **Gateway:** ether2
- **Distance:** 2

CLI

```
/ip address add address=203.76.120.10/30 interface=ether1

/ip dhcp-client add interface=ether2 add-default-route=no

/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
/ip firewall nat add chain=srcnat out-interface=ether2 action=masquerade
```

```
/ip route add gateway=203.76.120.9 distance=1
/ip route add gateway=ether2 distance=2
```

Topic 8.11 - Scenario 6: Static + Private

Winbox

Step 1: Static IP

- 203.76.120.10/30 → ether1

Step 2: Private IP

- 192.168.1.2/24 → ether2

Step 3: NAT

- ether1 → masquerade
- ether2 → masquerade

Step 4: Route

Primary:

- 203.76.120.9 → distance 1

Backup:

- 192.168.1.1 → distance 2

CLI

```
/ip address add address=203.76.120.10/30 interface=ether1
/ip address add address=192.168.1.2/24 interface=ether2

/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade
/ip firewall nat add chain=srcnat out-interface=ether2 action=masquerade

/ip route add gateway=203.76.120.9 distance=1
/ip route add gateway=192.168.1.1 distance=2
```

Topic 8.12 - Netwatch (Smart Failover Monitoring System)

Step 1: Netwatch কী?

1. **Netwatch = Router** দিয়ে **internet alive** আছে কিনা **check** করার **tool**
2. **Router** একটি **IP-**তে **ping** করে (যেমন): **ping 8.8.8.8**
3. **Result** অনুযায়ী **action** নেয়

Step 2: Netwatch কেন দরকার?

Problem (Without Netwatch)

- **PPPoE connected** ✔
- **Cable** ঠিক ✔
- কিন্তু **internet** নাই ❌

Router internet test করে না **so, Failover** হয় না

Step 3: Netwatch কীভাবে solve করে?

👉 **Netwatch Router-**কে বলে:

“শুধু **connection** না, **internet check** করো”

👉 **Router ping** দেয়:

8.8.8.8

👉 **Result**

Condition	Router Action
Ping OK	সব ঠিক
Ping FAIL	ISP down

👉 **Easy Analogy**

- **Router = guard**
- **Cable** = দরজা

দরজা খোলা মানে কি ভিতরে মানুষ আছে? না ❌, **Netwatch = guard** ভিতরে ঢুকে দেখে ✔

👉 **Final Clear Answer**

1. **Router default**ভাবে **internet check** করে না
2. শুধু **connection check** করে, **PPPoE connected ≠ Internet working**

3. **Router checks connection, Netwatch checks internet**

Step 4: Winbox Setup (Step-by-Step)

Winbox Setup

Netwatch Open

- **Winbox → Tools → Netwatch**

Add Entry

- **Click → +**

Basic Settings

Field	Value
Host	**8.8.8.8**
Interval	**5s**

Down Script (Internet Down)

```
/ip route disable [find comment="Primary"]
```

Primary route বন্ধ

Up Script (Internet Back)

```
/ip route enable [find comment="Primary"]
```

Primary route চালু

CLI Setup

```
/tool netwatch add host=8.8.8.8 interval=5s down-script="/ip route disable [find comment=Primary]" up-script="/ip route enable [find comment=Primary]"
```

Netwatch একবার সেট করলে **save** হয়ে যায়—প্রতিবার **Winbox-**এ ঢুকে আবার **command** দিতে হয় না।

Step 5: Important Concepts

Interval কী?

👉 কত সময় পরপর **check** করবে

5s = প্রতি ৫ সেকেন্ডে **ping**

Host কী?

👉 যে **IP-কে ping** করবে

✔ **Use: 8.8.8.8 or 1.1.1.1**

❌ **don't use google.com**

Script কী?

👉 **condition** হলে কী করবে

- **Down script** → **fail** হলে
- **Up script** → ঠিক হলে

Netwatch ব্যবহার করার সময় আমরা **script** দিয়ে **route enable/disable** করি।
এই **script-**এ **route select** করার জন্য `find` ব্যবহার করা হয়।

👉 কিন্তু এখানে ভুল **method** ব্যবহার করলে পুরো **network down** হয়ে যেতে পারে।

❌ **Wrong Method: find distance=1**
/ip route disable [find distance=1]

কী হয় এখানে?

👉 **Router** খুঁজে বের করে:

যেসব **route-**এর **distance = 1** → সবগুলো

Problem

ধরো **router-**এ আছে:

0.0.0.0/0 → PPPoE → distance=1   (Internet Route)
192.168.88.0/24 → bridge-lan → distance=1   (Local Route)

👉 এই **command** দিলে:

❌ **Internet route disable**
❌ **Local route-**ও **disable**

👉 **Result:**
পুরো **network down** হয়ে যেতে পারে

✅ **Correct Method: find comment="Primary"**
/ip route disable [find comment="Primary"]

কী হয় এখানে?

👉 **Router** শুধু খুঁজে বের করে:

comment="Primary" যেটা আছে → শুধু ওই **route**

👉 **Result:**

✔ শুধু **Primary Internet route disable**
✔ অন্য **route safe** থাকে

কেন **comment** ব্যবহার করবো?

Method	Result
distance	multiple route match হতে পারে
comment	exact route control করা যায়

Easy Analogy

- find distance=1 → "সব লাল শার্ট পরা লোককে ধরো"
- find comment="Primary" → "শুধু **Rahim**-কে ধরো"

Failover script = Use comment, not distance

Step 6: Check Netwatch Status

CLI

```
/tool netwatch print
```

👉 **status** দেখাবে:

up / down

Step 7: Common Mistakes

❌ **Comment** না দেওয়া

👉 **script** কাজ করবে না

❌ **Wrong host**

👉 **Internet down** হলেও **detect** হবে না

❌ **NAT** না দেওয়া

👉 **backup** কাজ করবে না

✅ **Easy Method:**

Failover = Comment + Correct IP + NAT

Hand Practical 1: Internet + Remote Access Router Setup

Topic 1.1 – Remote Access via Private IP

Network Diagram

ISP Router
10.250.46.229
|
|
ether1 (WAN)
10.250.46.230/30

MikroTik Router
192.168.88.1/24

ether2 (LAN)
|
Local Network

Step 1: Configure WAN IP
CLI:

```
/ip address add address=10.250.46.230/30 interface=ether1 comment=WAN
```

Winbox:

1. Open Winbox.
2. Go to IP → Addresses.
3. Click +.
4. Enter:
    - Address: 10.250.46.230/30
    - Interface: ether1
    - Comment: WAN
5. Click Apply and OK.

Expected Result

- ether1 receives the WAN IP.
- Router can communicate with upstream ISP router.

Step 2: Configure LAN IP
CLI:

```
/ip address add address=192.168.88.1/24 interface=ether2 comment=LAN
```

Winbox:

1. Go to IP → Addresses.
2. Click +.
3. Enter:
   - Address: 192.168.88.1/24
   - Interface: ether2
   - Comment: LAN
4. Click Apply and OK.

Expected Result

- LAN gateway becomes 192.168.88.1.

Step 3: Configure Default Route
CLI:

```
/ip route add gateway=10.250.46.229 comment=Default_Route
```

Winbox:

1. Go to IP → Routes.
2. Click +.
3. Enter:
   - Gateway: 10.250.46.229
   - Comment: Default_Route
4. Click Apply and OK.

Expected Result

- Internet traffic forwards to ISP gateway.

Step 4: Configure DNS
CLI:

```
/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes
```

Winbox:

1. Go to IP → DNS.
2. Set:
   - Servers: 8.8.8.8, 1.1.1.1
3. Enable:
   - Allow Remote Requests
4. Click Apply and OK.

Expected Result

- Router can resolve domain names.
- LAN users can use router as DNS.

Step 5: Configure NAT
CLI:

```
/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade comment=NAT_WAN
```

Winbox:

1. Go to IP → Firewall → NAT.
2. Click +.
3. Under General:
   - Chain: srcnat
   - Out Interface: ether1
4. Under Action:
   - Action: masquerade
5. Click Apply and OK.

Expected Result

- LAN users can access the internet.

Step 6: Firewall Configuration(Input Chain)

CLI

```
/ip firewall filter add chain=input connection-state=established,related action=accept comment="INPUT Allow Established/Related"
/ip firewall filter add chain=input connection-state=invalid action=drop comment="INPUT Drop Invalid"
/ip firewall filter add chain=input protocol=icmp action=accept comment="INPUT Allow Ping"
```

```
/ip firewall filter add chain=input protocol=tcp dst-port=8080 action=accept comment="INPUT Allow WebFig 8080"
/ip firewall filter add chain=input protocol=tcp dst-port=8989 action=accept comment="INPUT Allow Winbox 8989"
/ip firewall filter add chain=input protocol=tcp dst-port=22 action=accept comment="INPUT Allow SSH"

/ip firewall filter add chain=input in-interface=ether2 action=accept comment="INPUT Allow LAN Access"
/ip firewall filter add chain=input action=drop comment="INPUT Drop All Others”
```

Winbox:

1. Go to IP → Firewall → Filter Rules.
2. Create each rule one by one.
3. Use:
   - Chain: input
   - Protocol: tcp or icmp
   - Destination Port: 8080, 8989, 22
   - Action: accept
4. Add final drop rule:
   - Chain: input
   - Action: drop

Expected Result

- WebFig accessible on port 8080
- Winbox accessible on port 8989
- SSH accessible on port 22
- Unauthorized access blocked.

Step 7: Firewall Configuration(Forward Chain)

CLI

```
/ip firewall filter add chain=forward connection-state=established,related action=accept comment="FORWARD Allow Established/Related"
/ip firewall filter add chain=forward connection-state=invalid action=drop comment="FORWARD Drop Invalid"
/ip firewall filter add chain=forward in-interface=ether2 out-interface=ether1 action=accept comment="FORWARD LAN to Internet"
/ip firewall filter add chain=forward action=drop comment="FORWARD Drop All Others”
```

Winbox:

1. Go to IP → Firewall → Filter Rules.
2. Add rules:
    - Allow established/related
    - Drop invalid
    - Allow LAN to WAN forwarding
    - Drop all others

Expected Result

- LAN users can browse internet.
- Unauthorized forwarding blocked.

Step 8: Configure Services
CLI

```
/ip service disable telnet
/ip service disable ftp
/ip service disable api
/ip service disable api-ssl
/ip service disable www-ssl

/ip service set www disabled=no port=8080
/ip service set winbox disabled=no port=8989
/ip service set ssh disabled=no port=22
```

Winbox:

1. Go to IP → Services.
2. Disable:
    - Telnet
    - FTP
    - API
    - API-SSL
    - WWW-SSL
3. Change ports:
    - WWW → 8080
    - Winbox → 8989
    - SSH → 22

Expected Result

- Only required management services remain enabled.

- Router security improves.

Step 9: Remote Access Test

WebFig

http://10.250.46.230:8080

Winbox

10.250.46.230:8989

SSH

ssh admin@10.250.46.230

Topic 1.2 – Remote Access via Public IP

Network Diagram

Internet
|
Public Gateway
27.147.201.14
|
ether1 (WAN)
27.147.201.13/30
MikroTik Router
192.168.88.1/24
ether2 (LAN)
|
Local Network

Step 1: Configure WAN IP
CLI:

```
/ip address add address=27.147.201.13/30 interface=ether1 comment=WAN
```

Winbox:

1. Go to IP → Addresses.
2. Click +.
3. Enter:
   - Address: 27.147.201.13/30
   - Interface: ether1
4. Click Apply and OK.

Step 2: Configure LAN IP
CLI:

```
/ip address add address=192.168.88.1/24 interface=ether2 comment=LAN
```

Winbox:

5. Go to IP → Addresses.
6. Click +.
7. Enter:
   - Address: 192.168.88.1/24
   - Interface: ether2
   - Comment: LAN
8. Click Apply and OK.

Expected Result

- LAN gateway becomes 192.168.88.1.

Step 3: Configure Default Route
CLI:

```
/ip route add gateway=27.147.201.14 comment=Default_Route
```

Winbox:

1. Go to IP → Routes.
2. Add:
   - Gateway: 27.147.201.14

Step 4: Configure DNS
CLI:

```
/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes
```

Winbox:

Go to IP → DNS and configure DNS servers.

Step 5: Configure NAT
CLI:

```
/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade comment=NAT_WAN
```

Winbox:

6. Go to IP → Firewall → NAT.
7. Click +.
8. Under General:
    - Chain: srcnat
    - Out Interface: ether1
9. Under Action:
    - Action: masquerade
10. Click Apply and OK.

Expected Result

- LAN users can access the internet.

Step 6: Firewall Configuration(Input Chain)

CLI

```
/ip firewall filter add chain=input connection-state=established,related action=accept comment="INPUT Allow Established/Related"
/ip firewall filter add chain=input connection-state=invalid action=drop comment="INPUT Drop Invalid"
/ip firewall filter add chain=input protocol=icmp action=accept comment="INPUT Allow Ping"

/ip firewall filter add chain=input protocol=tcp dst-port=8080 action=accept comment="INPUT Allow WebFig 8080"
/ip firewall filter add chain=input protocol=tcp dst-port=8989 action=accept comment="INPUT Allow Winbox 8989"
/ip firewall filter add chain=input protocol=tcp dst-port=22 action=accept comment="INPUT Allow SSH"

/ip firewall filter add chain=input in-interface=ether2 action=accept comment="INPUT Allow LAN Access"
/ip firewall filter add chain=input action=drop comment="INPUT Drop All Others”
```

Winbox:

5. Go to IP → Firewall → Filter Rules.
6. Create each rule one by one.
7. Use:
   - Chain: input
   - Protocol: tcp or icmp
   - Destination Port: 8080, 8989, 22
   - Action: accept
8. Add final drop rule:
   - Chain: input
   - Action: drop

Expected Result

- WebFig accessible on port 8080
- Winbox accessible on port 8989
- SSH accessible on port 22
- Unauthorized access blocked.

Step 7: Firewall Configuration(Forward Chain)

CLI

```
/ip firewall filter add chain=forward connection-state=established,related action=accept
comment="FORWARD Allow Established/Related"
/ip firewall filter add chain=forward connection-state=invalid action=drop
comment="FORWARD Drop Invalid"
/ip firewall filter add chain=forward in-interface=ether2 out-interface=ether1
action=accept comment="FORWARD LAN to Internet"
/ip firewall filter add chain=forward action=drop comment="FORWARD Drop All Others”
```

Winbox:

3. Go to IP → Firewall → Filter Rules.
4. Add rules:
   - Allow established/related
   - Drop invalid
   - Allow LAN to WAN forwarding
   - Drop all others

Step 8: Configure Services
CLI

```
/ip service disable telnet
/ip service disable ftp
/ip service disable api
/ip service disable api-ssl
/ip service disable www-ssl

/ip service set www disabled=no port=8080
/ip service set winbox disabled=no port=8989
/ip service set ssh disabled=no port=22
```

Winbox:

4. Go to IP → Services.
5. Disable:
   - Telnet
   - FTP
   - API
   - API-SSL
   - WWW-SSL
6. Change ports:
   - WWW → 8080
   - Winbox → 8989
   - SSH → 22

Expected Result

- Only required management services remain enabled.

Step 9: Remote Access Test
WebFig

http://27.147.201.13:8080

Winbox

27.147.201.13:8989

SSH

ssh admin@27.147.201.13

Security Recommendations

1. Use strong passwords.
2. Change default admin username.
3. Restrict Winbox and SSH access using trusted IP addresses.

4. Keep RouterOS updated.
5. Disable unused services.
6. Use VPN for secure remote management.
7. Backup configuration regularly.

Topic 1.3 – Remote Access via PPPoE

Step 1: Configure LAN IP

CLI

```
/ip address
add address=192.168.88.1/24 interface=ether2 comment=LAN
```

Winbox

1. IP → Addresses
2. Click +
3. Enter:
   - Address: 192.168.88.1/24
   - Interface: ether2
   - Comment: LAN
4. Apply → OK

Step 2: Configure PPPoE Client

CLI

```
/interface pppoe-client
add name=pppoe-out1 interface=ether1 user=Indor password=123456 disabled=no
add-default-route=yes use-peer-dns=no
```

Winbox

1. PPP → Interfaces
2. Click + → PPPoE Client
3. Configure:
   - Interface: ether1
   - User: Indor
   - Password: 123456
   - Add Default Route: ✓
   - Use Peer DNS: ✗

4. Apply → OK

Step 3: Configure DNS

CLI

```
/ip dns
set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes
```

Winbox

1. IP → DNS
2. Servers:
   - 8.8.8.8
   - 1.1.1.1
3. Check:
   - Allow Remote Requests
4. Apply → OK

Step 4: Configure NAT

CLI

```
/ip firewall nat
add chain=srcnat out-interface=pppoe-out1 action=masquerade comment=NAT
```

Winbox

1. IP → Firewall → NAT
2. Click +
3. General:
   - Chain: srcnat
   - Out Interface: pppoe-out1
4. Action:
   - Action: masquerade
5. Apply → OK

Expected Result:

- LAN users can access the Internet.

Step 5: Configure Services

CLI

```
/ip service
disable telnet
disable ftp
disable api
disable api-ssl
disable www-ssl

set winbox port=8989 disabled=no
set www port=8080 disabled=no
set ssh port=22 disabled=no
```

Winbox

1. IP → Services

Disable:

- Telnet
- FTP
- API
- API-SSL
- WWW-SSL

Modify:

Service	Port
Winbox	8989
WWW	8080
SSH	22

Apply → OK

Step 6: Configure Firewall (Input Chain)

CLI

```
/ip firewall filter
add chain=input connection-state=established,related action=accept
comment="Established"

add chain=input connection-state=invalid action=drop comment="Drop Invalid"

add chain=input protocol=icmp action=accept comment="Allow Ping"

add chain=input protocol=tcp dst-port=8989 action=accept comment="Allow Winbox"

add chain=input protocol=tcp dst-port=8080 action=accept comment="Allow WebFig"

add chain=input protocol=tcp dst-port=22 action=accept comment="Allow SSH"

add chain=input in-interface=ether2 action=accept comment="Allow LAN"

add chain=input action=drop comment="Drop Others”
```

Winbox

IP → Firewall → Filter Rules

Create rules in this order:

Chain	Condition	Action
input	established,related	accept
input	invalid	drop
input	ICMP	accept
input	TCP Port 8989	accept
input	TCP Port 8080	accept
input	TCP Port 22	accept
input	In Interface=ether2	accept
input	all others	drop

Expected Result:

- Winbox accessible on port 8989
- WebFig accessible on port 8080
- SSH accessible on port 22

Step 7: Configure Firewall (Forward Chain)

CLI

```
/ip firewall filter

add chain=forward connection-state=established,related action=accept comment="Forward Established"

add chain=forward connection-state=invalid action=drop comment="Forward Invalid"

add chain=forward in-interface=ether2 out-interface=pppoe-out1 action=accept comment="LAN to Internet"

add chain=forward action=drop comment="Drop Others"
```

Winbox

IP → Firewall → Filter Rules

Create rules:

Chain	Condition	Action
forward	established,related	accept
forward	invalid	drop
forward	ether2 → pppoe-out1	accept
forward	all others	drop

Expected Result:

- LAN users can browse the Internet.
- Unauthorized forwarding blocked.

Step 8: Remote Access Test

যদি **PPPoE Public IP** হিসেবে 103.127.57.73 পায়, তাহলে:

WebFig

http://103.127.57.73:8080

Winbox

103.127.57.73:8989

SSH

ssh admin@103.127.57.73

Heading 9: VLAN Management (ISP Production Level)